University of Brasilia

Institute of Exact Sciences
Computer Science Department

Slot Allocation Problem: Medium hub airport

Lucas Kuniyoshi

Dissertation presented as a partial requirement for the qualification of the Professional Master's Degree in Applied Computing

Advisor
Prof. Dr. André Luiz Peron Martins Lanna

Brasilia
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Resumo

A alocação de slots das companhias aéreas dentro da infraestrutura aeroportuária influencia diretamente a lucratividade de um aeroporto e sua eficiência em absorver o crescente tráfego de transporte aéreo de passageiros~~(ACI World Insights, 2025)~~. Embora a expansão da capacidade aeroportuária represente a solução definitiva para o desequilíbrio entre esse crescimento e a infraestrutura limitada~~(IATA, 2025)~~, fatores como regulamentações, áreas delimitadas e limitações financeiras frequentemente exigem que os aeroportos implementem mecanismos alternativos de gestão da demanda. ~~Esta tese~~ Essa dissertação tem como objetivo apresentar um método para resolver o problema de alocação de slots aeroportuários, focando em um modelo que emprega Programação Inteira (*Integer Programming* – IP), Aprendizado de Máquinas (*Machine Learning* – ML) e Teoria das Filas (*Queueing Theory* – QT). «UNIFICAR ESSAS TRES FRASES SEGUINTES(SOBRE AS TÉCNICAS QT, ML E IP) EM UMA ÚNICA FRASE» Especificamente, QT é utilizada para simular a capacidade real de um aeroporto, identificando gargalos. ML, por sua vez, busca preencher lacunas de informação cruciais para um processo de alocação aprimorado, particularmente detalhes frequentemente retidos pelas companhias aéreas para projeções de longo prazo. IP tem sido amplamente adotada na literatura como um mecanismo de gestão de slots, levando à redução de atrasos e deslocamento de slots para as companhias aéreas. ~~Adicionalmente, uma Literatura Sistemática de Literatura () foi conduzida, investigando as abordagens existentes para o problema de alocação de slots. Este processo gerou um protocolo de revisão, dados de pesquisa e um relatório abrangente detalhando o conhecimento avaliado sob múltiplas perspectivas, as metodologias empregadas e as áreas identificadas para pesquisa futura. Consequentemente, esta tese oferece~~ Essa dissertação busca oferecer uma contribuição tríplice ~~, demonstrando uma aplicação unificada desses métodos para abordar o problema: (i) Uma abrangente do problema de~~ ao propor um método que integre as técnicas de QT, ML e IP para a alocação ~~de slots, incluindo um protocolo que garante a reprodutibilidade e validade da revisão, juntamente com uma síntese narrativa explorando as principais motivações, métodos, resultados e limitações do tópico por meio de mapeamento conceitual; (ii)~~ eficiente de *slots* em aeroportos, apresentar um estudo de caso ~~do~~ aplicado ao Aeroporto Internacional de Brasília (BSB) ~~, que utiliza~~

~~para avaliar a capacidade do aeroporto; (iii)~~ para avaliar sua capacidade de operação pelo emprego de QT e, por fim, propor uma alocação de submissões iniciais de ~~slots utilizando , cujo objetivo é minimizar o deslocamento de slots~~ *slots* solicitados pelas companhias aéreas ~~, com~~ pela utilização de IP baseado em informações faltantes obtidas por meio da aplicação de ML.

Palavras-chave: Problema de alocação de slots em aeroportos, Programação inteira, Aprendizado de máquina, Teoria das filas, Otimização

Abstract

The allocation of airline flight schedules within airport infrastructure directly influences an airport's profitability and its efficiency in absorbing growing passenger air transportation traffic (ACI World Insights, 2025). While expanding airport capacity represents the ultimate solution to the imbalance between this growth and the limited infrastructure (IATA, 2025), factors such as regulations, delimited areas, and financial limitations frequently require airports to implement alternative demand management mechanisms. This thesis aims to present a method for solving the airport slot allocation problem, focusing on a model employing Integer programming (IP), Machine Learning (ML), and Queueing Theory (QT). Specifically, QT is utilized to simulate an airport's actual capacity, identifying bottlenecks. ML, conversely, seeks to bridge information gaps crucial for an improved allocation process, particularly details often withheld by airlines for long-term projections. IP has been widely adopted in literature as a slot management mechanism, leading to reduced delays and slot displacement for airlines. Additionally, a Systematic Literature Review (SLR) was conducted, investigating existing approaches to the slot allocation problem. This process generated a review protocol, research data, and a comprehensive report detailing the assessed knowledge from multiple perspectives, the methodologies employed and identified areas for future research. Accordingly, this thesis offers a threefold contribution, demonstrating a unified application of these methods to address the problem: (i) A comprehensive SLR of the slot allocation problem, this includes a protocol ensuring the reproducibility and validity of the review, alongside a narrative synthesis exploring the topic's primary motivations, methods, results, and limitations through conceptual mapping; (ii) a case study of Brasilia International Airport (BSB), which utilizes QT to evaluate the airport's capacity, (iii) an allocation of initial slot submissions using IP, his allocation aims to minimize the displacement of slots requested by airlines, with missing information obtained through ML application.

Keywords: Airports' slot allocation problem, Integer programming, Machine learning, Queueing theory, Optimization

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Chapter 1

Introduction

1.1 The continuous growth of the air transportation industry

The air transportation industry has been growing since 1945. It passed through several economic and geopolitical crises, the most recent being the COVID-19 pandemic (ACI, 2025). The latter has significantly impacted the world passenger traffic, reducing the industry by 40%, when comparing 2021 against 2019, before the spread of the virus(KALIĆ et al., 2022). This diminish of the sector cost approximately 324 billion dollars of gross passenger operating revenue for airlines (HASEGAWA, 2022) The COVID-19 also influenced the airport's revenue and operations. At the beginning of the pandemic, the close sky policies allowed only essential flights, reducing the use of the airport's infrastructure. Sanitary measures also changed the aircraft movement itinerary, due to mandatory cleaning and inspections before and after transporting passengers, which directly affected the demand that could be accepted by the airports. Even through all these negative indicators, it is estimated a recovery of almost 103% of worldwide passenger transportation in 2024, when comparing against 2019 (IATA, 2022). Some regions, such as Central America, have projections that surpass the passenger demand of 2019 at the end of 2023. Therefore, even with this uncertainty, the increasing demand required by the airlines, to compose a better network, will result in congestion of several airports, thereat old discussions of how to efficiently resolve the distribution of aircraft operations at airports come back into evidence.

1.2 Airport Conception

In order to better understand this study's main problem, it is important to have knowledge of the basic concepts of airport's operation. The prototypical airport is composed of 2 parts: an (i) airside and (ii) landside (SHERRY, 2009). From an operational airport's perspective, the aircraft movement starts at the airside, when it approaches the runway through the commands of the Air Traffic Management (ATM). After the landing at the runway, the aircraft continues the movement, now at the ground level, through the taxiways. The aircraft approaches the apron in the direction of the parking stand. When it is positioned at its designated stand, the aircraft parks and bridges or airstairs are attached to the equipment in order to disembark the passengers, leading them to a gate. After the gate, the passengers are at the landside, in the airport terminal. There, they may enjoy the commercial stores (e.g., convenience stores, restaurants, and car rental agencies) before collecting their luggage and leaving the airport by ground transport or connecting to other flights.

The opposite movement is also possible: the passengers arrive at the airport using ground transportation, they may also use the airport's parking lots; they access the terminal, after check-in procedures and security inspections; they may enjoy the commercial stores, before embarking to the aircraft; at the airside, the aircraft pushes back, after it gets disattached from the airport bridges or airstairs; it proceeds through the taxiways until it arrives at the runways; and finally it takes-off, ending its movement at the airport.

With the prototypical airport conception, it is possible to start the discussion about two main concepts: (i) how the airport generates revenue, and (ii) what are the main limitations of the airport infrastructure. These two are main components of the airport's slots allocation problem.

1.3 Airport Revenue Generation

The airport revenue is mainly generated through aeronautical and non-aeronautical sources, both of which are directly related to the aircraft traffic volume. The former is composed by: the fare paid by each passenger using the airport infrastructure, usually included in the flight cost paid to the airlines; and the movement charge, which is paid by the airline when it lands, uses the infrastructure (e.g., hangar and terminal), and takes-off from the airport. The latter is the revenue generated by: the commercial concession at the terminal, such as rental fees for the retail at the airport area; the real estate exploitation, which includes hotels, restaurants, and parking lots at the airport perimeter; and services offered to passengers and airlines. A market equilibrium between the two revenues sources is needed, since reduction of the aeronautical charges commonly means stimulation to the non-aeronautical revenues, through passenger demand increments, but it does not guarantee the a overall maximum return (ICAO, 2013).

In Europe (2019), it is shown by ACI (2021) that 54% of the overall airport revenue was generated by aeronautical sources. ICAO (2013) points out that 63% of the aeronautical income comes from charges directly applied to passengers. The article also argue that airport users do not pay the full cost of the infrastructure the use, explaining that, in 2012, 69% of the airports worldwide were loss-making. Seeking to profitability, the airports have increased the variety of commercial activities. It can cushioning the impact of lower passenger and freight volumes, due to its higher profit margins. This creates a cross-subsidization between aeronautical and non-aeronautical revenues sources, since profits from commercial exploitation, for example, can be reinvested in airport infrastructure, increase the demand of passenger or aircraft that can be accepted at the airport, reducing capital needs and overall cost.

Knowing that airport are important for a country social-economic-growth and, as it was said, with an efficient management it can achieve profitability, governments have decided that, under the right economic conditions, they can included private sector participation (e.g., outright ownership, short or long-term concessions, public-private-partnership schemes, and management contracts) for the financing and operation of an airport infrastructure. The airports, under this new ownership model, have sought for better ways to compete for both passenger and airlines. The aviation community agrees with the need of infrastructure investments to accommodate the growth of the industry, increasing the profit margins. Since most of the infrastructure added are in large increments, and the time needed for these ventures is extensive, the airports are exposed to considerable risk (ICAO, 2013). Even with the constant pressure for the airport expansion, most of the administrator have to better manage its current infrastructure, until the investment risks are reduced to an acceptable level.

1.4 Airport's infrastructure capacity

The limitation of the airport's infrastructure use is related with (ARC, 2021; IATA, 2020b):

1. How many aircraft movements the airport's runways can accept in a certain period (e.g., one hour, 30 minutes, 15 minutes, and 5 minutes);
2. How many stands are available at the airport, and what are their category (e.g., cargo or passenger commercial flights);

3. How many gates are available for the passenger movements;
4. How many passengers can be processed through the security inspection at the check-in;
5. How many passengers are acceptable for the terminal area available; and
6. How many aircraft can use the infrastructure through time slot and environmental restrictions.

The most restrictive infrastructure capacity can be used to accept movements at the airport, or a combination of several limitations can be best suited for the task to distribute slots without damaging the airport's level of service, such as movement delays and congestion at the taxiways.

1.5 Problem statement

The present study introduces two main reasons to elevate the efficiency in the initial slot allocation in an airport. The first reason is related to the airport's ways to generate revenue. The two main ways of an airport generates revenue (i.e., aeronautical and non-aeronautical) are directly related to the passenger traffic volume. In (GRAHAM, 2009), for example, it is discussed the importance of the commercial revenues, which represented (2017) 52.9% of the total airport revenue to Africa and the Middle East, 45.7% to Asia and Pacific, 38.1% to Europe, 52.6% to North America, and only 29.0% to Latin America and the Caribbean. The article blames the small exploitation of airports' commercial areas in Latin America and the Caribbean on the lack of airport traffic stimulus, which results in the full dependence of the regulated airport fares, and non-development in both commercial and aeronautic areas. In the Economic Commission for America Latin and the Caribbean (PLANZER; PÉREZ, 2019), it is demonstrated the growth of the worldwide air transportation market, before the COVID-19 pandemic (2017), which, by International Civil Aviation Organization (ICAO), would reach 10 billion passengers per year by 2040. Despite the great projection, the report also describes the Latin American and the Caribbean challenges to build airport infrastructure, the major impediment to the development of the local sector. The International Air Transport Association (IATA) has urged the local aviation authorities to ensure, between several highlighted topics, the efficient use of the available airport's capacity in the region. When the expansion of infrastructure it is no possible, the optimization of the allocation in the available slots is strongly sought by the airport administrator, to improve the revenue generation prospects. Some regions have already implemented more restrictive on-time performance goals for airlines (RAVIZZA et al., 2014), which aims to enhance aircraft's taxi times, take-off sequencing, and allocating push-back time, thus affecting the demand of movements that an airport can accept.

The second reason is related to the impact of the operational performance improvement in the airlines. As it was mentioned, the operational capacity of an airport, i.e., the ability to absorb the demand of flights, is directly related to its infrastructure (e.g., number of runways, gates, aircraft stands, and terminal areas). Although the mentioned features of an airport can be used to determine its capacity, a lower limit is set to mitigate the negative impacts of the airport's main partners, the airlines. This measure goes in the opposite direction of the first reason. When the limit is set greater than it should be, operational impacts are raised (e.g., delays). Most airlines interconnected their flights in a network. Therefore, a delay, for example, does not only impact the airline locally, but it can also spread its repercussions through other airports, impacting several aircraft itineraries and elevating the airline's operational costs. This topic is discussed in (BALL et al., 2010), where it is described that the airline's costs of crew management, disrupted passenger accommodation and aircraft re-position, due to delays, can reach 20 billion dollars a year (2017). Another negative impact is related to intense and not planned aircraft taxi traffic. In (NIKOLERIS; GUPTA; KISTLER, 2011), it is reported that the fuel consumed in the stop-and-go situations, resulting primarily from the congestion on the airport's taxiways system, can be approximately 18% of the fuel used during the whole operation.

For that reason, while both airports and air operators want to expand their operations, a balance between accepted demand, which increases revenues (for airports), and operational efficiency, which mitigates high operational costs (for airlines), is needed.

As it is highlighted in (CAVUSOGLU; MACARIO, 2021), there are three predominant processes for the airports' capacity allocation: (i) administrative management, with slot distribution processes through a set of specific prioritization rules; (ii) economic management, where the airport's congestion price is established, slot sales and exchanges are allowed, or auctions are made; and (iii) no regulatory mechanism, relying only on some infrastructure constraints as parameters (e.g., LaGuardia Airport and Ronald Reagan Washington National Airport as presented by (FAA, 2000)).

The administrative management is the most used mechanism for slots. It is often use the Worldwide Airport Slot Guidelines (WASG), published by IATA(RIBEIRO et al., 2018). There are also applications of WASG variations through local resolutions, as is the case of Brazil, where the local regulator is theNational Civil Aviation Agency of Brazil (ANAC). Therefore, this chapter will: (i) present administrative mechanisms, through the WASG and its allocation practices in congested airports, discussing the differences and similarities between the resolution covering Brazilian airports and the best practices developed by IATA; (ii) briefly explain the characteristics of the economic management, which contains market-driven mechanisms; and (iii) review advantages of slot management mechanism applications compared to airports with almost no restrictions.

1.6 Administrative demand management

Demand management has been demonstrated in several works as one of the main processes to distribute air operators' movements in the airports' available infrastructure, dealing with the lack of capacity, delays and operational service level decaying due to congestion (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; RIBEIRO et al., 2018).

The administrative demand management approach aims to increase airport efficiency with allocation prioritization. Based upon optimization techniques, such practices have as objective to allocate in a certain number of available resources (airport slots), the dependent activities, i.e., air operators' arrivals and departures movements. The allocation, made through the distribution of slots, must satisfy specific conditions (explained in 1.4) (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017).

At the strategic level, the administrative management develops a cooperation practice where actions are coordinated between stakeholders, with bilateral communications and frequent reviews, so that the allocation may take place in accordance with all parties. The general objective is to match the requests made by the air operators with the declared infrastructure capacity, maximizing the number of accepted movements and, at the same time, minimizing the air operators' operational cost to when they need to reschedule its flights outside the planned hours, due to the restrictions imposed at the airport (GILLEN; JACQUILLAT; ODONI, 2016). As it was explained, the main administrative management strategy is the IATA's WASG, which has been adopted by local authorities in several countries (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; FAIRBROTHER; ZOGRAFOS, 2018; CAVUSOGLU; MACARIO, 2021).

1.6.1 Worldwide Airport Slot Guidelines (WASG) - IATA

To provide a standard for slot management and operations planning, the IATA, together with Airports Council International (ACI) and Worldwide Airport Coordinators Group (WWACG), publishes the WASG(IATA, 2020b).

In general terms, the proposal to coordinate airports through the WASG aims to: (i) improve the connectivity of air services, promoting competitiveness in congested airports; (ii) provide slot schedules that meet demand and are consistent across seasons; (iii) ensure a fair, transparent and non-discriminatory slots allocation; (vi) promote efficient use of airports' infrastructure; (v) guarantee market access to new air operators; (vi) provide a flexible slot management system for the industry, which can be adapt through local regulations and market conditions; and (vii) minimize congestion and delays caused by the allocation of slots with no parameters.

Slot Stakeholders

Slot management impacts several groups that are interested in the growing demand and the efficiency of the airport infrastructure use. Among them, the WASG highlights:

1. The air operators, the origin of the movement demand in the airports;
2. The airport administrator, responsible for balancing the number of slots that will be made available and the adequate operational level;
3. The authorities responsible for airspace control;
4. Coordinators or facilitators responsible for managing slots at the airport; and
5. The government responsible for the airport use regulation.

The Figure 1.1 illustrates the relationship of stakeholders, showing the information that is exchanged during the slot management process.

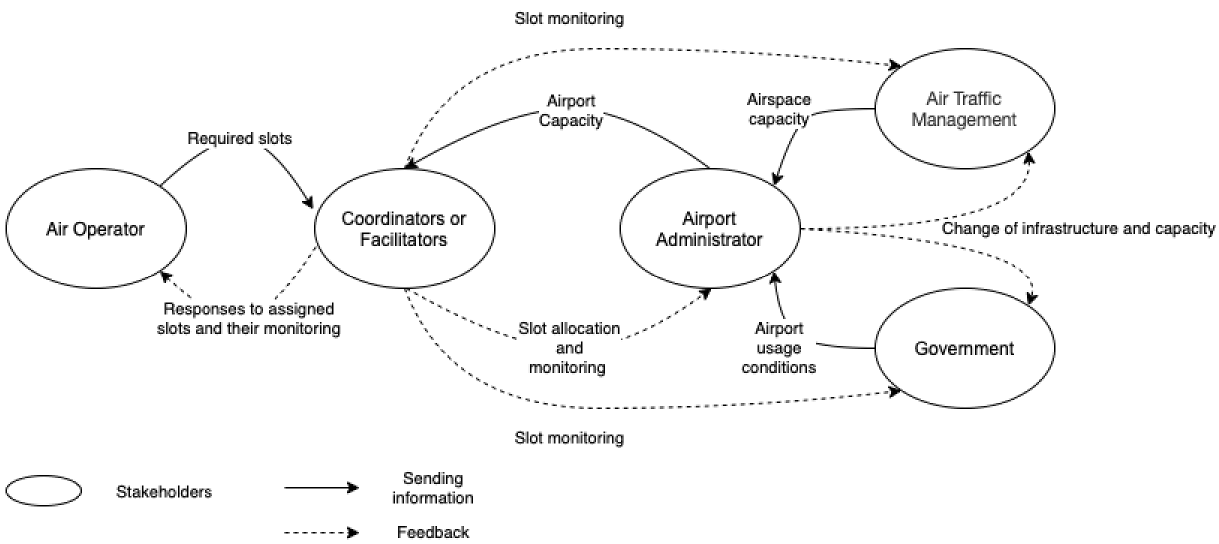

Figure 1.1: Relationship between stakeholders in the management of slots through the direct use of WASG.

Air operators have the responsibility to send requests for desired slots through a message, detailing the operations, and to prepare to cases where congestion is present in the airport and offers with different slots are sent as a response to the request. They also must follow the calendar of coordination activities and local regulations, monitoring the slots allocated to always mitigate the risk of airport infrastructure misuse.

The coordinator or facilitator are responsible to allocate the air operators slots in a transparent and non-discriminatory way. They must ensure that the allocation is made in accordance with the capacity declared by the airport and the priority criteria established in the WASG. During the activities, the they assist the understanding of the coordination parameters, local guidelines, regulations, and any other criteria used in the allocation. They must make available the list of the slots allocated and the offers sent, explaining why the requests were not fulfilled. They must attend the conferences, in order to better approach the needs of the air operators, trying to find a middle ground between competitors. They also must perform slot monitoring, to mitigate infrastructure misuse and to apply the use-or-lose-it rules, creating the historical slots.

The airport administrator must provide support to the facilitator or coordinator, assisting in their tasks. Airports must provide the infrastructure information so that capacity is filled with adequate demand. If there is any special limitation present in their capacity, airports must communicate with the other stakeholders, especially if the demand projection shows to exceed the available capacity in future seasons. Capacity updates must be presented at least twice a year, following each season's activity calendar. The airport must send the monitoring of the slots to the airspace authorities and local regulators.

The authorities responsible for air control must communicate to the airport administrator the airspace capacity that modifies the airport infrastructure capacity parameters. The government that concedes the airport, must, through regulations, send the airport operator the conditions for using the infrastructure, as well as inform the investments that the airport operator needs to do in order to meet the projected target demand.

Capability Analysis

Prior to the slot allocation, a demand and capacity analysis is performed. This analysis must be carried out promptly for the declaration of capacity, one of the season's official activities. It takes into account the airport's ability to absorb movement demand, together with airspace limitations, in a level that the quantity of movements accepted in the airport infrastructure does not degrade its service levels, i.e., does not promote congestion and delays. The result is constraints regarding the runway(s), stand(s), terminal(s), gate(s), among many other relevant airport infrastructures.

Another important limitation of capacity is environmental, i.e., the airport discusses the need to reduce the number of slots available when there is a relevant environmental impact (e.g., periods without movement due to noise pollution or due to environmental events that affects the aircrafts' visual operations).

One of the objectives of the capability statement is to understand which level the airport is at, i.e., Level 1, Level 2, and Level 3. The level indicates the the airport's coordination criticality.

In order to better understand the the slot management process, Table 1.1 shows a glossary containing the main terms used by WASG.

Table 1.1: Glossary of relevant terms in WASG's slot management (IATA, 2020b).

#	**Term**	**Definition**
1	Slots	A permission that is given by a coordinator for a planned operation to use an airport's infrastructure at a specific date and time.
2	Airport Level	There are three airport levels: (1) for those whose demand does not exceed the infrastructure capacity; (2) for those whose demand causes certain congestion at specific times (e.g., hours, days, days of week), but which can be resolved through mutual agreements between the facilitator and air operators; and (3) for those whose demand exceeds the airport capacity limits, requiring a coordinator for the slots allocation, using WASG's best practices and prioritizing allocations for better market competitiveness.
3	Infrastructure capacity and airport parameters	Compiled of the necessary parameters for the coordination of slots. Through these, it is possible to identify the operational capacity for allocation that does not exceed the demand limit, providing an adequate service level (e.g., maximum operations per hour / 30 min. / 15 min. / 5 min. on the runway, number of aircraft parking positions available on the apron, number of available gates, airspace limitations, environmental limitations, etc.).

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Table 1.1 - Continued from previous page.

#	Term	Definition
4	Seasons	There are two specific periods where operations take place. The (1) Summer Season, which starts on the last Sunday of March, and the Winter Season, which starts on the last Sunday in October.
5	Series of slots	A minimum of 5 slots allocated for approximately the same time on the same day of the week through a season.
6	Slot pool	All slots that will receive allocation priority at level 3 airports, after the historical slots are properly allocated.
7	Historic Slots	Slots with operating precedence at the allocated airport, acquired by a regularity above 80% in the equivalent previous season.
8	Facilitator	The one responsible for collecting data and adjusting movement at level 2 airports.
9	Coordinator	The one responsible for data collection and coordination of slots at level 3 airports.
10	Activity Calendar	Deadlines and Events that manage the process of coordinating movement and slots for each season. It is established two per year.
11	Previous Equivalent Season	Last Season of the same name, i.e., if the current season is a summer season, the previous equivalent season is the prior summer season.
12	New entrants	Air operators that have a small number of operations (less than 7) for each day of a season, or companies that do not operate yet and are requesting operations.
13	Annual movements	Are all movements that have specific times in both Summer and Winter seasons.

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Table 1.1 - Continued from previous page.

#	Term	Definition
14	Messages	Reference to data shared between coordinator, facilitator and air operators (IATA, 2020a). The messages are usually standardized in a format recognized by the industry (e.g., SSIM), containing critical information about the slot, i.e., operations' start date, operations' end date, air operator IATA's code, equipment type (e.g., Boeing 737 MAX), number of seats offered, type of movement air (e.g., Passenger or Cargo), frequency of operation during the week, hours of operation (slots), among others.

Airport level definition

At Level 1, due to the availability of adequate schedules for every demand, the allocation becomes free of priorities.

At Level 2, the following allocation priority is applied: (i) movements that operated regularly in the previous equivalent season; (ii) annual movements, i.e., those which have operated with high regularity in the immediately preceding season; (iii) movements with more schedule operations for the current season; (iv) temporary operations; (v) movements restricted by operational factors; and (vi) any other type of movement. At Level 2, as well as a Level 3, a waiting list can be instituted for already congested periods.

For Level 3, WASG proposes a mechanism that grants access to the high-demand market, opening opportunities for new operators to obtain slots at coveted times. For these, the allocation is made prioritizing a series of slots, temporary slots, and other operations. The series of slots can be allocated prioritizing: (i) historical slots with no changes, with changes other than in coordination parameters, or with changes that impact the coordination parameters within a flexibility range period; (ii) slot pool, containing a consistent division of the remaining slots for new entrants and non-new-entrants operators; and (iii) other movements. It should be noted that additional allocation factors are commonly applied to the Level 3 slot pool, such as annual movements within a flexibility range, movements with more operations for the current season, movements that are restricted to certain schedules due to regulation, movements with longer time at waiting lists, movements that have a specific type of service (e.g., passengers and cargo) or market (e.g., regional, long-distance and international destinations), movements with the greater number of frequencies, among other factors established by a stakeholders' committee.

Each airport level has a different allocation priority. The Figure 1.2 illustrates the allocations by airport level, specifying the prioritized movements.

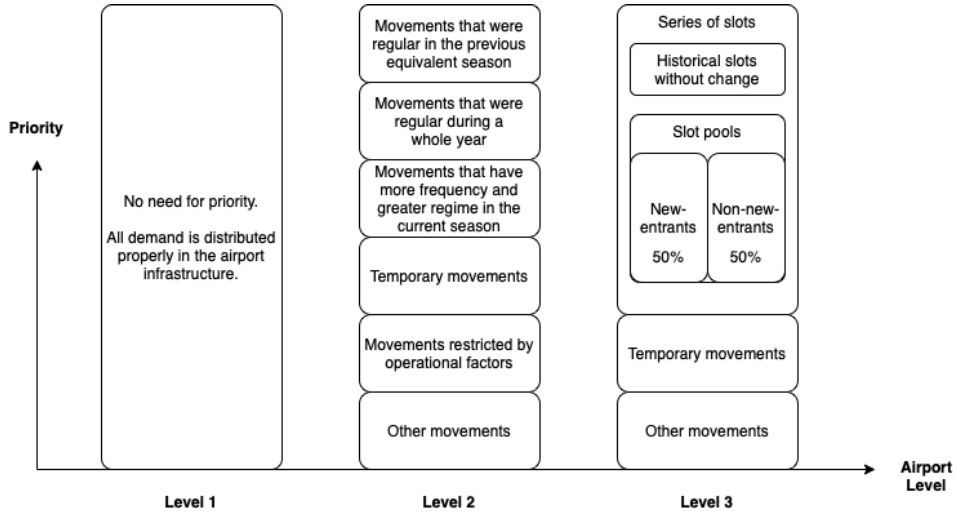


Figure 1.2: Allocation priority for each airport level.

Slot allocation: calendar activities

The slot allocation process, defined by the WASG activity calendar, has the following steps:

1. The calendar is published by the regulator for the immediately following season;
2. Airports must declare their capacity and coordination parameters within 7 days of initial submission;
3. Coordinators must provide details of historic slots through standard messages (SHL);
4. Air operators must review the historic slots received (AHD), contacting the coordinators if they disagree with the SHL, which results in an open communication between stakeholders;
5. The air operators send the initial submission with the requested slots (ISD);
6. The coordinators inform the air operators of the result of the initial submission, i.e., the airport's network after the distribution of demand (SAL);
7. Coordinators and air operators enter into dialogue to adjust offers for more advantageous slots, which can be done in a conference (SC), where exchanges of slots

can occur between air operators, when local regulations allow them and when it is confirmed the advantage by the coordinator;

8. After the conferences, a continuous reallocation process begins until the limit for the return of slots (SRD), where any movement that will not be operated must be canceled;

9. With the reinstate of non-operational slots, the reference base (BDR) is created on which the 80% regularity will be required for historic status for the next equivalent season; and

10. The last step takes place through operational planning, where the airline community plans the airspace, stands, and gates, among others infrastructure need to be prepared to receive the movements.

After all the steps, new slots can be requested if feasible allocation. The season starts with the first planning operation on the last Sunday of October (Winter Season) or on the last Sunday of March (Winter Season). Simultaneously, the planning for the immediately following season takes place. The Figure 1.3 illustrates the season system, with each season indicated by specific colors. Note that during the activities of a specific season, the activities related to the immediately following season are already started, in a continuous process.

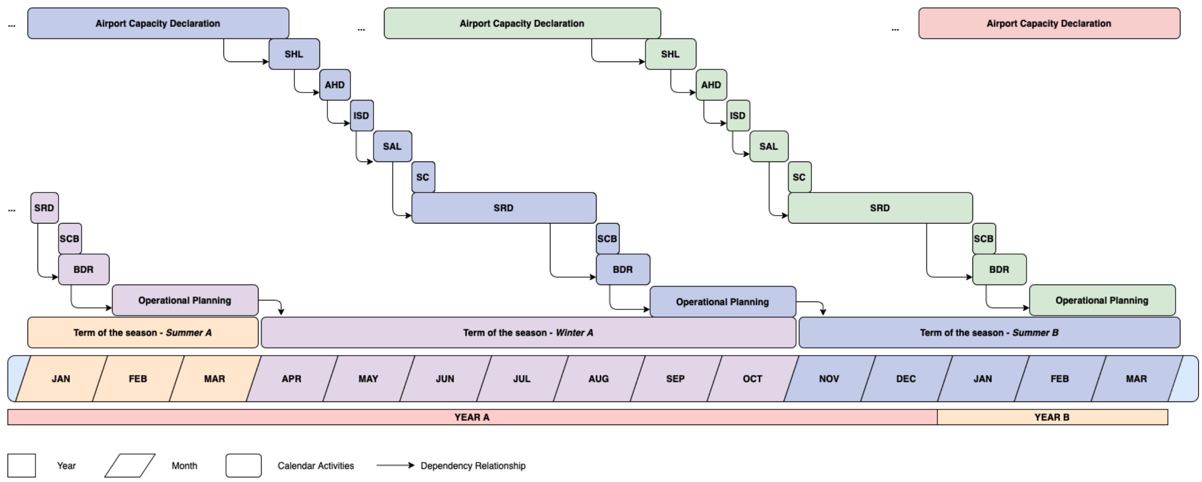


Figure 1.3: Illustration of the calendar of activities stipulated by the WSGA.

Slot Monitoring

For the slots monitoring, WASG recommends that it should be divided into (i) pre-operation analysis, which identifies the slots misuses before the day when the movements occur; and (ii) post-operation analysis, which identifies whether a misuse has occurred and whether the operator has achieved the historic right in its slots. The objective is to ensure proper operation, avoid the waste of airport infrastructure, and open communication channels between stakeholders.

1.6.2 Local Regulations: Brazil

As recommended by WASG, the application of good practices must consider the regulations in place where the airport is located. Thus, as the present work studies airports in Brazil, it is necessary to understand the rules established by the ANAC, highlighting their similarities and differences.

ANAC establishes, through Resolution No. 682, of July 07, 2022, the regulations and procedures for allocating slots at airports coordinated and facilitated. As established in WASG, coordinated airports would be Level 3 airports, and airports of interest would be level 2 airports, with their specificities explained in resolution (ANAC, 2022).

The principles of slot coordination are stipulated to minimize the effects of infrastructure congestion at Brazilian airports, focusing on transparency, non-discrimination, impartiality, and efficient use of the declared infrastructure of each airport. It also establishes an activity calendar containing the declaration of capacity, distribution of slot requests, allocation responses, and slots monitoring.

Unlike WASG, the resolution established an ANAC team that will be responsible for activities related to slot coordination at the level 3 airports. The Figure 1.4 illustrates the relationship between stakeholders via ANAC regulation. Thus, if the level of congestion at a given Brazilian airport exceeds the limit of its infrastructure capacity to absorb demand, in any of its allocation parameters (e.g., runway, patio, or terminal), or at due to the request of an interested party, ANAC can declare it as coordinated. In this case, although the agency is also the government agency responsible for monitoring the development of the airport exploration plan, a separate ANAC team will also be in charge of the administrative management of slots.

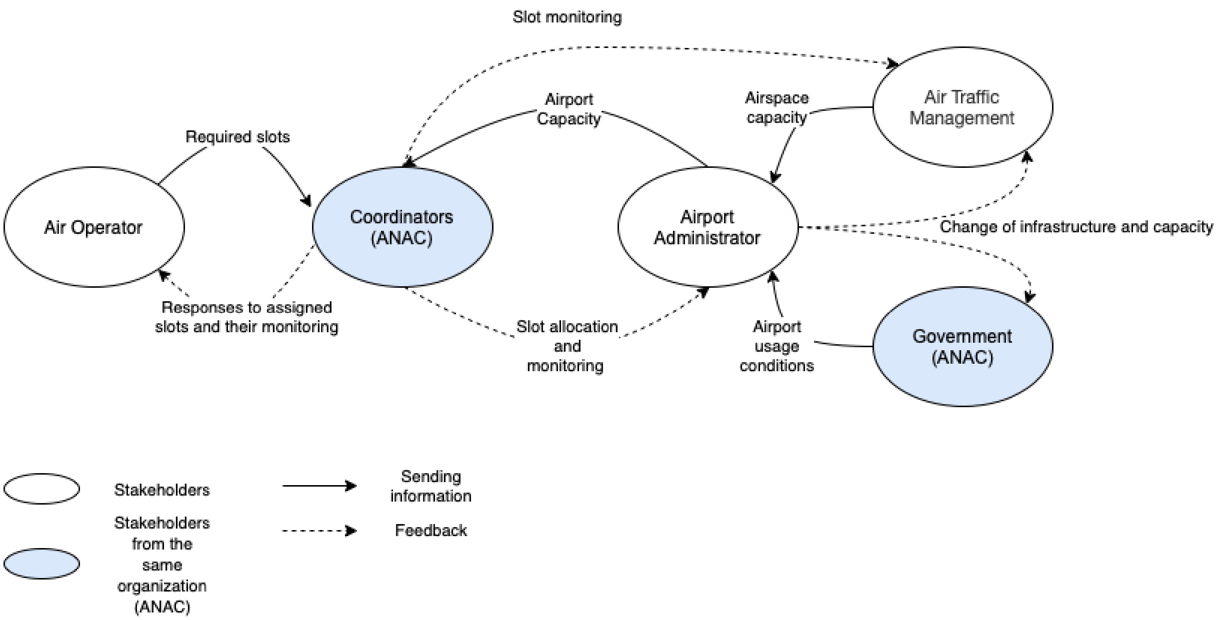


Figure 1.4: Relationship between stakeholders in the management of slots through the use of Brazilian resolution (level 3 airports).

Another relevant difference in the distribution of slots in Brazil is that during the initial submission ISD only requests that constitute a series of slots are processed, making temporary requests impossible. The order of priority also differs from WASG, starting with (i) historic slots, (ii) historic slots that have changed, and (iii) new requests (respecting the division of the slots pool). Being the last one allocated in SAL, prioritizing the continuation of the immediately previous season followed by the new operations.

Some criteria, in addition to those mentioned above, can subsidize priority when there is a tie or conflict in slot allocation (not in order of priority): (i) the number of operations during the season; (ii) the largest aircraft, i.e., the number of seats or by the amount of cargo that can be transported; (iii) the operational efficiency in the previous equivalent season; (iv) best environmental efficiency; and (v) requests that best promote the airport competition, i.e., reduction of the Herfindahl-Hirschman Index (HHI).

For the analysis of operational efficiency, the resolution presents a comparison of the reference base BDR with the current base of slots for the season, taking into account the operations that did not use the available infrastructure without valid justification. The Regularity Index (IR) is calculated using the Equation 1.1. The IR will be used to determine the historic status of the slots for the next equivalent season.

$$IRs = \frac{Used\ slots}{Slots\ allocated\ in\ the\ BDR} \tag{1.1}$$

Similar to WASG, if the company falls below the minimum regularity target (in this case established by the coordinator), as well as when the intentional misuse of allocated slots is identified, the company may lose the historic slot. In the ANAC's regulation, when misuse happens, if the slots are requested at the next equivalent season, they would be inserted in the slots pool.

ANAC also mitigates the misuse of slots through regulations on infractions and their respective administrative measures. If an air operator obtains a slot that it does not intend to use, wasting the airport infrastructure above the imposed limit, it will be considered an infraction punishable by fines established by the agency, as well as the impediment to obtain historic slots.

In Brazil, it is set to facilitated airports (i.e., level 2 airports) to prioritize the allocation of air operators' slots series that operated in the last season (i.e., the annual movements). Since there are no historic slots, the resolution recommends lower priority for operators who misuse airport infrastructure. The Figure 1.5 illustrates the stakeholders in an facilitated airport configuration stipulated by ANAC. Note that both the airport and the slot facilitator are the same entity.

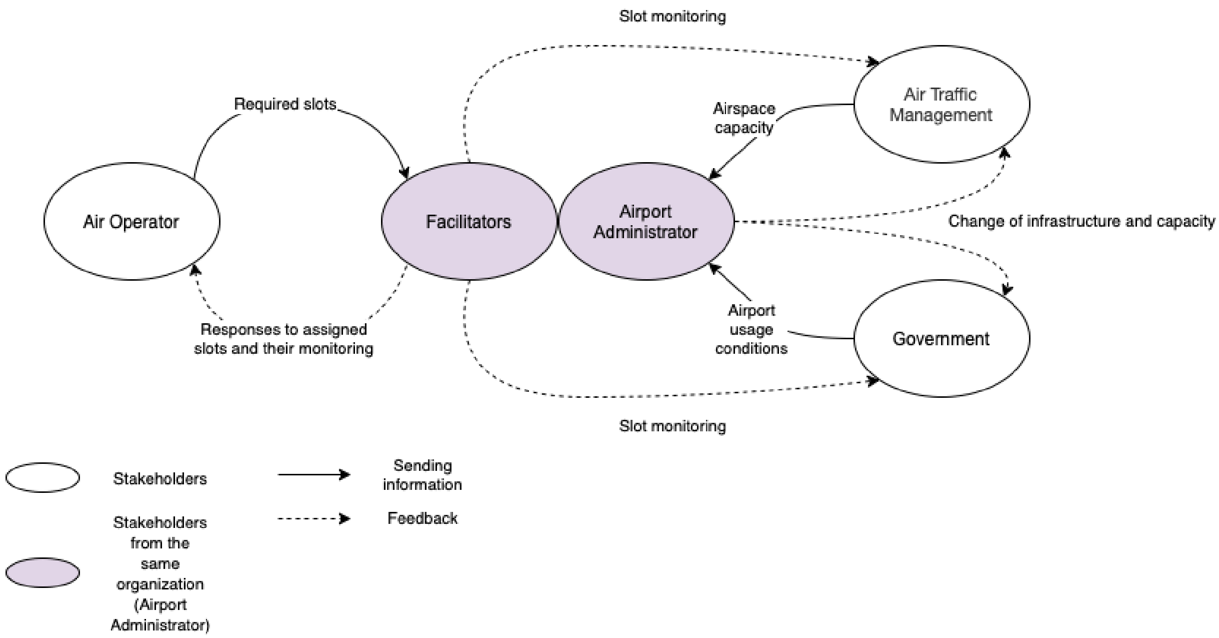


Figure 1.5: Relationship between stakeholders in the management of slots through the use of Brazilian resolution (level 2 airports).

1.7 Economic demand management

Although it is not the scope of the present study, research with market-driven approaches, i.e., economic, has been developed, despite not being widely applied in practice, to solve the problem of increasing demand allocation in airports. The main objective is to address challenges not solved by the administrative mechanisms (BALL; DONOHUE; HOFFMAN, 2006).

The article Gillen, Jacquillat & Odoni (2016) cites two primary mechanisms for the economic management of the exposed problem: (i) Congestion pricing, where the cost of using the overloaded infrastructure is passed on to the air operators, which are interested to gain access to the airport; and (ii) auction of slots, as well as in the administrative management of demand, the airport declares the number of slots available through airport parameters, and it is given access to the air operator with the highest monetary offer.

1.7.1 Congestion pricing

The Figure 1.6 demonstrates the idea of price mechanisms, illustrating the relationship between the marginal cost of delay and the marginal benefit of allocation (GILLEN; JACQUILLAT; ODONI, 2016). Therefore, when the congestion price is not present, air operators can allocate “A” movements, resulting in a marginal delay cost “W”. However, the total marginal cost, “Y”, exceeds the allocation benefit. Reducing the amount of the number of moves allocated to “B” $\leq$ “A”, the total marginal cost is equal to the benefit of allocation “Z”, having an internal marginal cost “X”. There are, however, some difficulties with this demonstration, as (GILLEN; JACQUILLAT; ODONI, 2016) points out: (i) developing the congestion pricing scheme is complex, i.e., measuring the internal and total marginal cost is not trivial; and (ii) there are uncertainties regarding how the allocation of air operators varies with the congestion price.

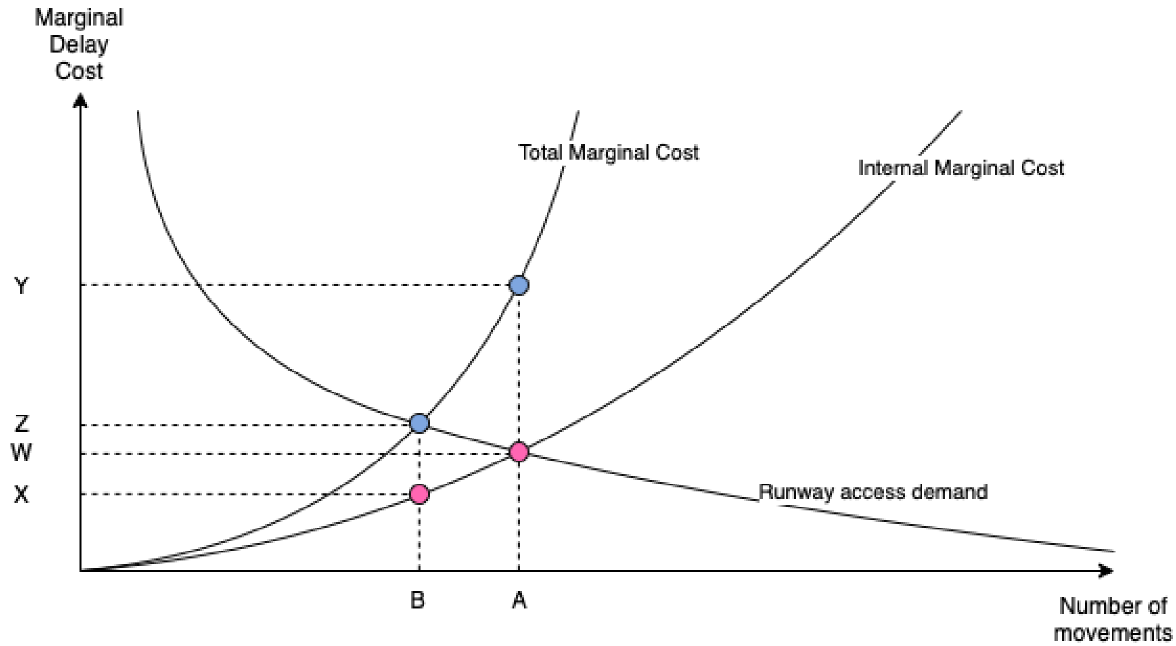


Figure 1.6: Congestion pricing as an alternative for external delay correction and slot allocation. Adapted from: (GILLEN; JACQUILLAT; ODONI, 2016).

To better understand the problem of economic management, it is important to point out that typically air operators allocate several movements in a short period, imposing congestion costs on themselves. It can be argued that airport systems containing hubs of few air operators, i.e., reduced competition, have a high internalization of congestion cost, reducing the price per slot, and making impacts related to economic demand management negligible (BALL; DONOHUE; HOFFMAN, 2006; GILLEN; JACQUILLAT; ODONI, 2016). However, for these airports, where the internal marginal cost is equal to the total marginal cost due to the lack of competition, it is shown higher rates of delays since overschedule are made by the monopolist air operator, that allocates demand which the operational cost cannot be totally internalized (BALL; DONOHUE; HOFFMAN, 2006).

The Figure 1.7 demonstrates the relationship between competition and increased cost per slots. In a non-competitive airport system, the cost of delay is internalized and there are no incentives to manage demand. However, introducing competition, it makes the cost per slots increase, distribution mechanisms that search for a balance between accepted moves and total delay cost are needed.

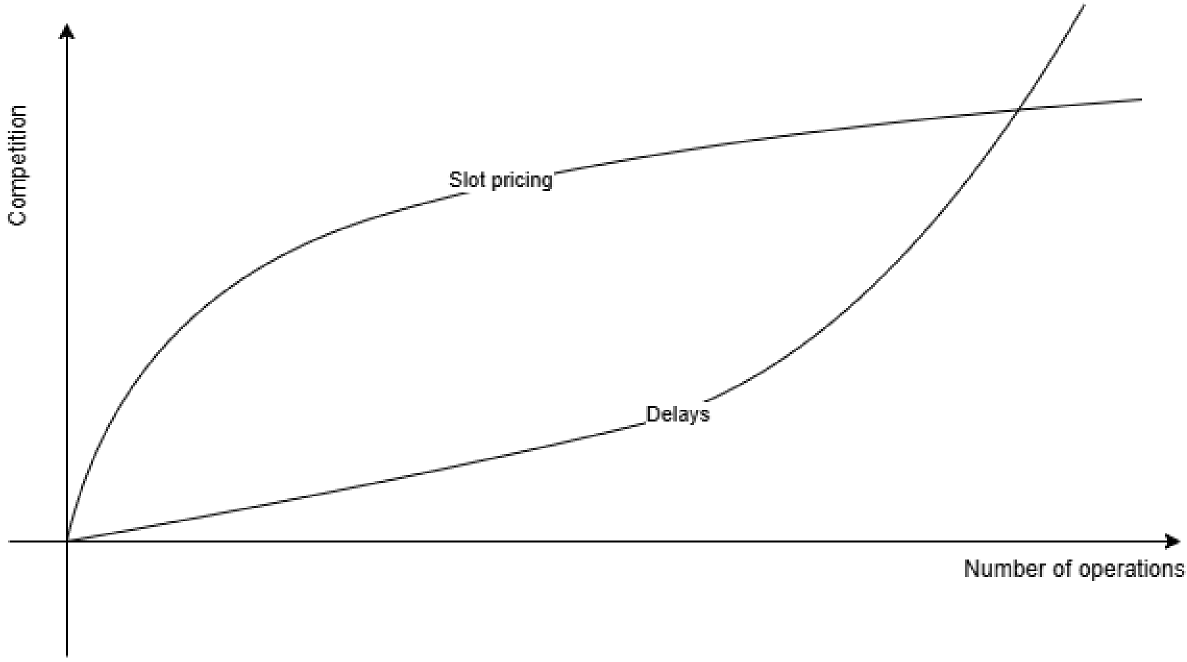


Figure 1.7: Relationship between the price of slots and costs related to competition. Adapted from: (BALL; DONOHUE; HOFFMAN, 2006).

Since congestion pricing is a topic that still requires further studies, due to its complexity in measuring the internalization of delays and designing the congestion pricing scheme for each modality of air operator (GILLEN; JACQUILLAT; ODONI, 2016), other economic management mechanisms would be the auctions of slots. These would also not be atomized, i.e., each slot auctioned individually, due to the risk of exposure to the air operator if arrivals and departures at a given airport do not end up coupled with each other and among other movements at other airports. This brings the need for combinatorial auctions, which need to be mutually exclusive.

1.7.2 Auction of slots

The knowledge that the slot will be auctioned in the future brings the continuous need to evaluate the market and the slots' price, limiting the negative sides of the administrative management of slots: (i) inefficiencies guaranteed by historic; (ii) legal difficulties in congested airports to appoint holders of newly released slots; and (iii) limitation of market entry for new air operators, as they have a less economic impact on the airport. It is worth mentioning that the revenue generated by the auction is related to the market value due to the scarce system, paying for the cost of the used infrastructure. However, a slot auction system should focus primarily on encouraging competition, rather than maximizing revenue (BALL; DONOHUE; HOFFMAN, 2006).

For the auctions mechanism, (BALL; DONOHUE; HOFFMAN, 2006) points out limitations in 5 pillars:

- ***Capacity***: the number of slots available for auction is related to demand, which is unknown before the auction takes place. Therefore, before declaring slot holders, it is necessary to ensure that the combination of slots granted does not exceed airport parameters.
- ***Ownership***: it is necessary to specify the period of slots ownership, together with their acceptable operating range, which also influences the capacity. Regulation that allows secondary auctions, where ownership over the slot can be transferred between operators, is also needed. Daily transfers between operators may also be possible, allowing better use of airport infrastructure.
- ***Value***: although the value of a slot is present to air operators, its value during an auction is unknown. Therefore, the market will evaluate and pay the slot holder an amount that will be corrected over time.
- ***Market Power***: it is necessary to design auctions that have a regulation that encourages competition, preventing offers to buy a monopoly.
- ***Implementation***: even if the transition from administrative management to an economic management system is organized in phases, i.e., auctioning a percentage of the slots in a congested airport, it will impact operators' connection networks airlines as a whole, and their losses to the market may be greater than the benefits.

A hybrid use is possible, i.e., combining the advantages of administrative management with economic management. Thus, the allocation could first come from an administrative allocation process (e.g., WASG and adaptations made by the local authorities), selling at congestion price or auctioning the remaining slots. This would add up to three important benefits of administrative mechanisms: (i) it helps air operators to project future decisions related to slots; (ii) it makes the process less complex by following the activities calendar, which proposes allocations twice a year; and (iii) it assists in the long-term industry projection, which can easily predict the growth of competition and the market, with the incentive of congestion pricing and slot appreciation, offered by economic management. However, it is worth mentioning that the disadvantages of both management would also be aggregated, where regulatory needs for slot distribution, ownership definition, and market access would be evident, and could be disputed in court by airline operators, as occurred in 2008 in an attempt to auction 10% of New York airports' slots (CAVUSOGLU; MACARIO, 2021; GILLEN; JACQUILLAT; ODONI, 2016; BICHLER et al., 2021).

1.8 Airports free of demand management mechanisms

As well as economic demand management, the approach to airports with systems free of demand management mechanisms goes beyond the scope of this work, but it has been applied in American airports and its cost-benefits have been highlighted (CAVUSOGLU; MACARIO, 2021; GILLEN; JACQUILLAT; ODONI, 2016).

There are two objectives to be achieved at airports in terms of demand and slots: (i) maximize the number of movements accepted in the airport infrastructure, adapting required schedules to possible operating hours; and (ii) minimize congestion-related costs, which are present in high-demand airports in different forms (e.g., delays) (CAVUSOGLU; MACARIO, 2021). It is noted that the objectives are inversely proportional because as the number of movements in an airport increases, projecting growth in revenue, the delays and impacts for the air operators increase, as illustrated in the Figure 1.8 (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; SWAROOP et al., 2012; GILLEN; JACQUILLAT; ODONI, 2016).

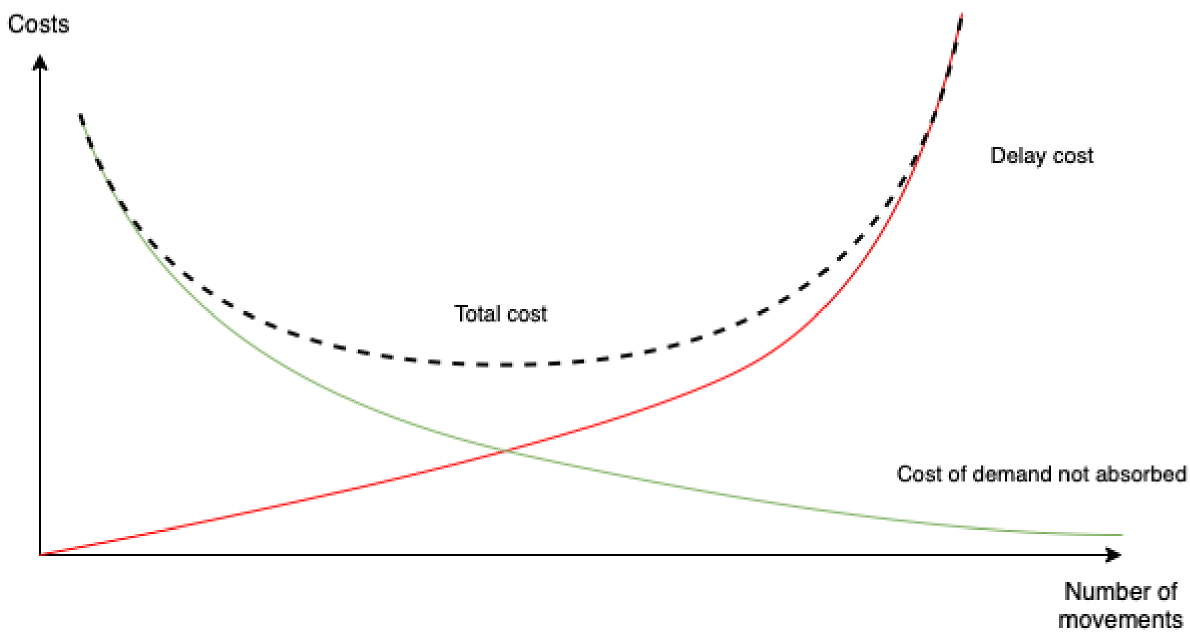


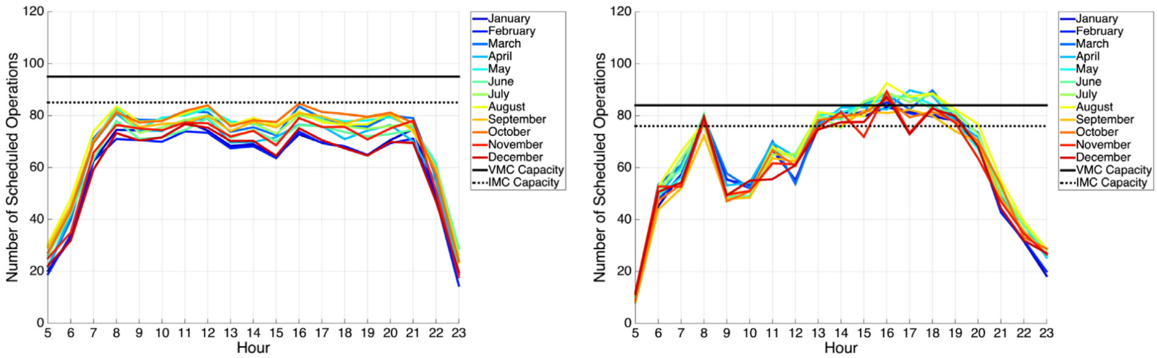
Figure 1.8: Cost-benefit of accepting movements in congested airports. Adapted from: (SWAROOP et al., 2012; GILLEN; JACQUILLAT; ODONI, 2016).

Although airports with high movement acceptance, and without demand management, are the simplest solution for allocation distribution, taking better advantage of the airport infrastructure (GILLEN; JACQUILLAT; ODONI, 2016), the approach also has risks and disadvantages.

The indiscriminate use of infrastructure helps generate revenue from stakeholders, who invest in the growth of the industry. In (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017) the authors argue that demand management often constrains the declared capacity of the airport, which in turn affects not only the slots accepted but the distance between the desired times of movements, i.e., the quality of slots in the airport in an air operator perspective. Thus, reducing management provides a greater frequency of movement, competition between operators, opportunities for new entrants, and better use of scarce infrastructure. On the other hand, the opening would also reflect on delay costs and slot reliability.

In (SWAROOP et al., 2012) the cost-benefit of applying demand management in airports with few mechanisms (e.g., LGA) was studied. It brings a methodology to simulate the cost related to increased movement and the costs generated by delays. Adding the benefits of 16 US airports in the simulation, the reduction in delay costs for passengers would be more than 200 million dollars annually. The study also proposes a reduction in infrastructure capacity, i.e., fewer movements during congested hours, resulting in a projection greater than 600 million dollars annually in benefits for passengers. However, the authors make it clear that, although the use of demand management mechanisms can generate financial benefits and a better level of service for passengers, by excluding movements, they could seriously harm the network of affected air operators. The costs generated with the implementation of these mechanisms were also not added. Although, intuitively, reducing delay costs also reduce operating costs, consequently the price of tickets for passengers.

Another important topic brought up by demand management mechanisms is capacity declaration. When an airport without these mechanisms is compared with coordinated airports, such as Frankfurt Airport (FRA) and Newark Liberty International Airport (EWR), which have the similar infrastructure, Figure 1.9, there is a concentration of movements in EWR, while FRA has a more balanced distribution. This characteristic is common in uncoordinated airports, where there are periods which scheduled movement exceeds the recommended limit of demand absorption, even in Instrumental Meteorological Conditions (IFR), i.e., adverse weather conditions of operation. In EWR, there is overutilization in the night period and underutilization in the rest of the day, showing inefficiency in the allocation. This factor affects the airport's punctuality performance, which, due to the non-linear relationship between the number of movements and delays when the airport operates close to its capacity limit, makes certain American airports perform worse when compared to equivalents in Europe (GILLEN; JACQUILLAT; ODONI, 2016).



(a) FRA (b) EWR

Figure 1.9: Average weekly movements per month for EWR and FRA airports in 2007. Source: (GILLEN; JACQUILLAT; ODONI, 2016).

The article (CAVUSOGLU; MACARIO, 2021) also points out that airports subject to coordination mechanisms tend to use the Visual Meteorological Conditions (VMC) operational limit of the runway as their maximum capacity, i.e., normal weather conditions. This is due to infrastructure overload due to adverse weather when delays are frequent. Operating above this conditional, as shown in the 1.9 part (b), even in VMC, already configures acceptance of excessive delays and low confidence in slot schedules, which would worsen in situations of IFR. In addition, the study (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017) points out that common concentrations in small time slots also increase the operational delay. By reducing infrastructure capacity, movements would be spread over new slots, minimizing the airport's overall delay. The Figure 1.10 illustrates the movement of the largest airport in Greece, where a reduction from 85% (part a) to 71% (part b) has a considerable impact on the airport's peak delays.

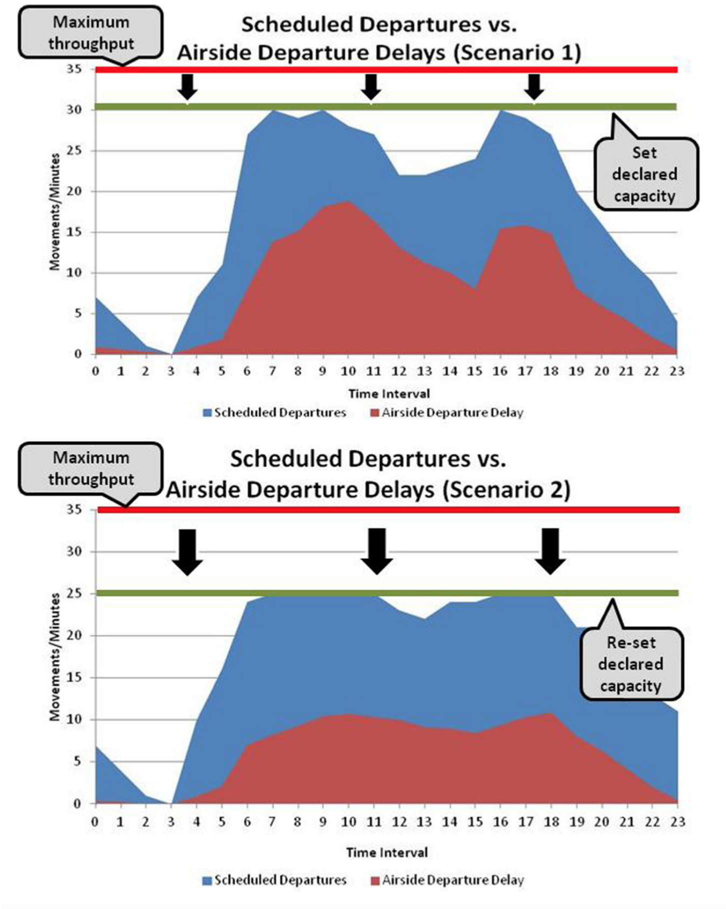


Figure 1.10: Modeling the effects of capacity declaration on congestion-related delays. Source: (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017).

Although there are difficulties in demand management systems (e.g., inefficiencies in the general use of infrastructure guaranteed by history, market opening, capacity declaration, and implementation) its mechanism provides a better definition of infrastructure parameters, which as consequence provides better demand absorption at adequate service levels (MORISSET; ODONI, 2011).

Chapter 2

State-of-the-art: Sistematic Literature Review

2.1 Planning the review

To gain a comprehensive understanding of the main problem and existing approaches, a Systematic Literature Review (SLR) was conducted following a transparent process to minimize bias and enhance reliability.

The SLR adhered to a three-step methodology: (i) Planning, this stage involved defining the research need, translating research questions into specific search terms, and developing a detailed review protocol prior to data collection; (ii) Conducting the Review, this stage encompassed identifying relevant research addressing the slot allocation problem, selecting primary studies, assessing their quality, extracting key data, and synthesizing the findings; and (iii) Presenting the Review Results, this stage aimed to summarize the key research findings related to the central problem and demonstrate how the proposed methods in this paper can address identified gaps in the existing literature.

A thorough understanding of airport slot allocation necessitates a comprehensive literature review to identify, analyze, and interpret diverse perspectives on the problem and its existing solutions. This review provides historical context and clarifies the potential contributions of this paper. To ensure a bias-mitigated and transparent data collection process, the review criteria and procedures were meticulously planned before execution, adhering to the guidelines outlined in Wohlin et al. (2012) and Shamseer et al. (2015). Figure 2.1 illustrates the sequential processes and their respective deliverables.

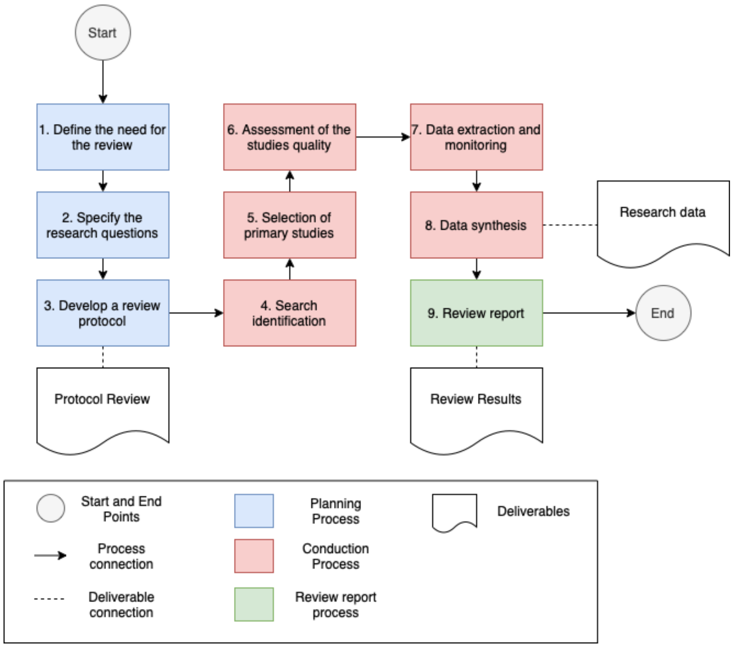


Figure 2.1: The literature review processes and their deliverables. Adapted from: (WOHLIN et al., 2012).

2.1.1 The need for a review

The first step was the identification of the need for a review, i.e., understand how the problem has been addressed in past research and what are the previous limitation.

Airport infrastructure exhibits significant variability, encompassing distinct configurations of runways, taxiways, gates, and terminals. Furthermore, airports operate within diverse environments, including densely populated areas, which significantly influences demand. These unique characteristics can lead to diverse perspectives on the slot allocation problem and consequently, a range of proposed solutions. Given the well-established nature of the slot allocation problem, a thorough understanding of its historical management is crucial.

While a *snowballing* approach could potentially gather relevant literature, it carries the risk of introducing search bias, potentially limiting the comprehensiveness and reproducibility of the review (JALALI; WOHLIN, 2012). A systematic review offers a more rigorous approach, ensuring a comprehensive understanding of the problem through a well-defined, step-by-step protocol that is subject to review and potential improvement.

This review assumed the absence of prior systematic reviews on this specific topic. However, if any such reviews were encountered, their contributions and limitations would be carefully evaluated

2.1.2 The SLR researches questions

The second step involved formulating specific research questions to guide the literature search. These research questions serve as crucial focal points for identifying relevant primary studies and extracting and analyzing relevant information. To ensure clarity and focus, the PICO framework was employed to systematically break down each research question into its essential components: Population, Intervention, Comparison, and Outcome.

Furthermore, the 'Intervention' component was specifically defined to clarify the approach of the present paper and determine whether similar approaches have been previously utilized in the literature.

The following research questions were formulated for this literature review:

- **RQ1:** In medium and large airports (P), how do integer programming, queue theory, and machine learning techniques (I) compared to the IATA's guidelines for the administrative management of slots (C) in terms of their effectiveness in reducing the number of changes to airlines' initial allocations and optimizing airport infrastructure utilization (O)?

To further refine the inquiry, the following three sub-questions were formulated: (i) What methodologies have been employed to solve the airport slot allocation problem? (ii) What are the primary constraints associated with these approaches? (iii) Are there any identified gaps or underlying assumptions that warrant further exploration? It aims to collect the most valuable information about the models created to solve the problem and how experimentation with algorithms are helping them to tackle it.

2.1.3 Review protocol

The third and final step in the planning phase involved the development of a comprehensive review protocol. This protocol serves as the foundation for the current research review and plays a crucial role in ensuring the reproducibility and validity of the study. The following key elements were considered during the development of the review protocol:

Search strategy for primary studies

The primary activities within this step involved:

- **Formulating a comprehensive search string:** this was achieved by translating the research questions into a set of relevant keywords.

- **Executing database searches:** the formulated search string was then applied to a range of relevant academic databases.

The search string was constructed using a combination of keywords, including: "Airport", "Slot", "Allocation", "Integer Programming", "Queue Theory", "Machine Learning", "Optimization", and "IATA Slot Guidelines".

To refine the search, boolean operators were employed to effectively combine these keywords, resulting in the following search string:

(airport **AND** "slot allocation" **AND** optimization **AND** ("IATA guidelines" **OR** "Worldwide Airport Slot Guidelines") **AND** ("machine learning" **OR** "integer programming" **OR** "queue theory")) **AND** [E-Publication Date: (01/01/2002 **TO** 01/12/2023)].*

Search sources

The formulated search string was subsequently employed to systematically search the following academic databases to identify relevant primary research studies:

1. ACM Digital Library;
2. Bielefeld Academic Search Engine (BASE);
3. EBSCO;
4. ScienceDirect;
5. Scopus;
6. Transport Research International Documentation (TRID);
7. Web of Science;
8. Digital Bibliography & Library Project (DBLP);
9. Directory of Open Access Journals (DOAJ);
10. IEEE Xplore;
11. JSTOR; and
12. Springer Link (Article section).

The selection of databases was guided by two primary criteria: (i) Multidisciplinary Coverage, the databases were chosen to encompass both multidisciplinary subjects and those specializing in computer science or transportation studies, aligning with the core research areas of this paper; and (ii) Search Capability, the databases were selected based on their comprehensive coverage and robust search capabilities, as outlined in Kitchenham & Charters (2007), this selection criterion prioritizes databases that facilitate systematic searches through the utilization of queries and filters, ultimately ensuring the reproducibility of the search process with high recall and precision.

Following the criteria established by Gusenbauer & Haddaway (2020), databases were classified as 'primary' if they met all necessary quality requirements, including the possibility of query refinement, string size criteria, server response criteria, use of boolean operators, functionalities and reproducibility. Databases 1 to 7 were designated as primary databases, while the remaining databases were considered supplementary. No additional search strategies (e.g., searching specific journals and conference proceedings or contacting researchers to obtain unpublished material or grey literature) were employed.

This step necessitates the meticulous removal of all identifiable false positives and duplicate entries across the searched databases.

To validate the search string's effectiveness, the article by Ribeiro et al. (2018) was used as a benchmark. This article, previously identified as relevant to the research scope, must be retrieved within the search results from at least one of the targeted databases. Failure to retrieve this article necessitates a reformulation of the search string.

Study selection criteria

To mitigate bias, specific inclusion and exclusion criteria were established prior to the screening process. Employing the "First Pass" approach as established by Keshav (2007), all primary papers were screened by carefully reviewing their titles, abstracts, introductions, and conclusions

Inclusion criteria:

1. The study must address any aspect of the airport slot allocation problem, encompassing models, perspectives, constraints, methods, technical considerations, etc.; and
2. The study must propose a solution for the airport slot allocation problem utilizing integer programming, queue theory, or machine learning techniques.

Study exclusion criteria

Exclusion criteria:

1. **Duplicates and Old Versions**, articles are excluded if they are duplicates of other included articles or represent older versions of the same study. If an article appears

in multiple primary and/or secondary databases, it is retained in the database listed first in the "Search Sources" section and removed from the remaining databases.

2. **Publication Date**, articles published before 2001 are excluded. This timeframe is chosen to focus on research conducted after the significant market regulations implemented following the September 11th attacks.

3. **Scope of Slot Guidelines**, articles introducing slot allocation guidelines beyond those outlined in the IATA's Worldwide Slot Guidelines (WASG) (IATA, 2020b) are excluded.

4. **Focus on Auction-based Solutions**, articles that primarily focus on auction-based economic approaches to solving the airport slot allocation problem are excluded.

5. **Lack of Full-Text Access**, articles that are not accessible in full-text format are excluded.

6. **Irrelevant Scope**, articles that are not directly relevant to the research questions and do not address the core slot allocation problem, but instead focus on adjacent areas such as ground delays or weather impacts on airports, are excluded.

7. **Specific Context**, articles that focus on specific contexts, such as the September 11th attacks or the COVID-19 pandemic, are excluded to maintain a broader and more generalizable perspective.

8. **Language**, articles not published in English or Portuguese are excluded.

Study selection procedures

The removal of all irrelevant papers was conducted through a thorough assessment of each study against the established selection and exclusion criteria. This process involved a careful review of titles, abstracts, and conclusions, with a focus on identifying and excluding papers that did not meet the inclusion criteria. If more than one section of the paper (title, abstract, or conclusion) was deemed irrelevant to the scope of this review, the paper was excluded, and a concise justification for its exclusion was recorded.

Narrative Synthesis

Given the anticipated variability and potential differences among the included studies – stemming from the application of diverse methodologies across varied samples (e.g., different airports) and potentially leading to methodological heterogeneity - this review will employ a narrative synthesis approach to synthesize the extracted data and assess the overall body of evidence (CAMPBELL et al., 2018; CAMPBELL et al., 2019).

This method is particularly suitable when the heterogeneity of the included studies precludes statistical meta-analysis (POPAY et al., 2006; CAMPBELL et al., 2018; CAMPBELL et al., 2019).

Drawing upon the guidance for Narrative Synthesis outlined by Popay et al. (2006), this study will undertake the following key stages:

1. **Developing a theoretical understanding of the problem:** This stage will involve constructing an initial framework or theory of how the identified software problem is conceptualized and addressed within the existing literature, considering the different approaches and contexts.

2. **Developing a preliminary synthesis of findings through tabulation:** A systematic tabulation of key characteristics and findings from each included study will be created. This will facilitate an initial overview and comparison of the evidence.

3. **Exploring relationships within the data using concept mapping:** Concept mapping techniques will be employed to visually represent and explore the relationships between different concepts, interventions, outcomes, and contextual factors identified across the studies. This will aid in identifying patterns and potential associations.

4. **Assessing the robustness of the synthesis through critical reflection:** The final stage will involve a critical discussion of the synthesis process, considering the strengths and limitations of the included studies (as determined by the quality assessment), the potential for bias, and the overall confidence in the synthesized findings. This reflection will explicitly address the impact of methodological heterogeneity on the conclusions drawn.

2.2 Conducting the review

This section presents the outcomes of the systematic search and selection process, including a comprehensive review of the validity checks implemented throughout the study. These checks encompassed data extraction procedures and a rigorous quality assessment of the included studies.

2.2.1 Identification of research

The initial search yielded 31 sources across 12 databases. After removing 9 duplicates (29.03%), a total of 22 unique sources remained for further analysis, as detailed in Table 2.1.

Table 2.1: Search results per database

#	**DATABASE**	**SOURCES FOUND**
1	ACM Digital Library	4
2	Bielefeld Academic Search Engine (BASE)	2
3	EBSCO	3
4	ScienceDirect	13
5	Scopus	0
6	Transport Research International Documentation (TRID)	0
7	Web of Science	0
8	Digital Bibliography & Library Project (DBLP)	0
9	Directory of Open Access Journals (DOAJ)	0
10	IEEE Xplore	0
11	JSTOR	0
12	Springer Link	0
	TOTAL	22

All studies were identified within the designated primary databases, with ScienceDirect emerging as the most prolific source, contributing 59.09% of the total retrieved articles. Notably, three databases – DBLP, DOAJ, and IEEE Xplore – did not yield any results in the initial search, even when considering duplicates.

To ensure a comprehensive understanding of the relevant literature, a rigorous screening process was implemented. Following the "first pass" approach outlined by Wohlin et al. (2012), each study's title and abstract were independently and randomly reviewed. Studies were excluded if they did not directly address the airport slot allocation problem or failed to meet the established inclusion criteria. This screening process resulted in the selection of 19 primary studies, representing 86.36% of the unique sources identified.

For the exclusion criteria:

1. One article was not accessible;
2. One article was reviewing the slot allocation problem through economic approaches; and
3. One article was focusing in adjacent area (flight planning optimization).

Following the selection procedures, the systematic literature review (SLR) yielded 19 primary studies. Notably, 13.63% of the initial search results were identified as false positives, i.e., studies retrieved by the search string but ultimately deemed irrelevant to the research scope. Table 2.2 provides a summary of the primary studies identified per database, including the number of false positives observed after duplicate removal.

Table 2.2: Primary studies per database and the false positive results.

#	DATABASE	PRIMARY STUDIES	FALSE POSITIVES
1	ACM Digital Library	03	25.00%
2	Bielefeld Academic Search Engine (BASE)	02	00.00%
3	EBSCO	02	33.33%
4	ScienceDirect	12	07.69%
5	Scopus	00	-
6	Transport Research International Documentation (TRID)	00	-
7	Web of Science	00	-
8	CiteSeerX	00	-
9	Digital Bibliography & Library Project (DBLP)	00	-
10	Directory of Open Access Journals (DOAJ)	00	-
11	IEEE Xplore	00	-
12	JSTOR	00	-
13	Springer Link	00	-
	TOTAL	19	13.63%

The distribution of studies by year of publication is visualized in Figure 2.2. Notably, the subject of this investigation has gained significant attention in recent years, with 43.37% of the selected sources published between 2020 and 2023. The selection process is illustrated in Figure 2.3.

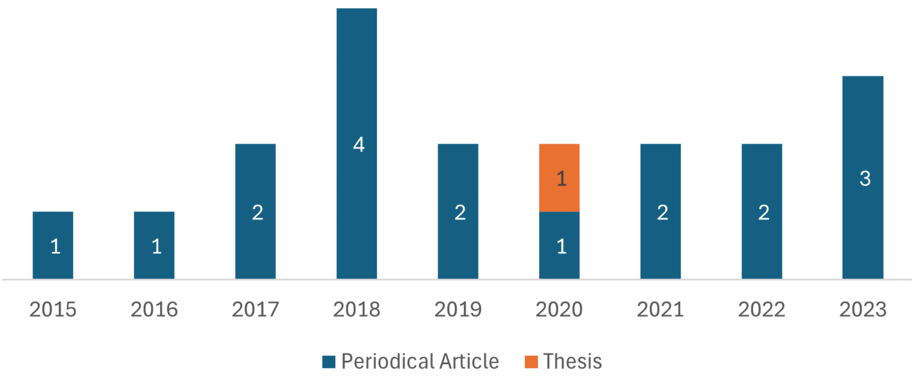


Figure 2.2: Number of studies by year of their publication.

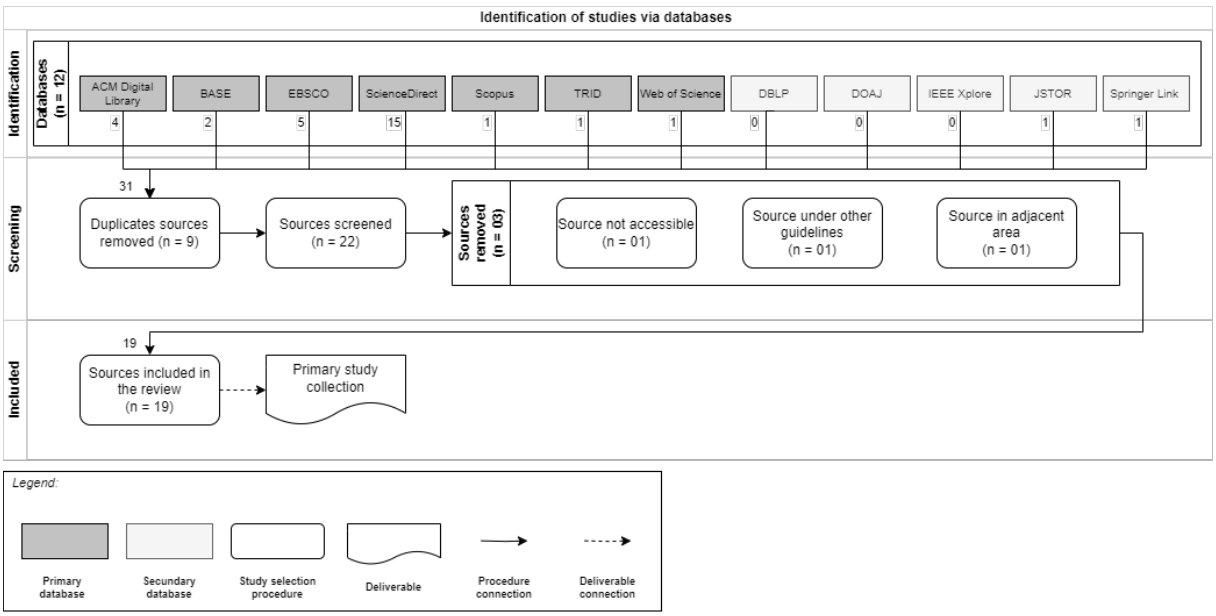


Figure 2.3: The studies selection procedures and deliverable. Adapted from: (PAGE et al., 2021).

2.3 Review report process

2.3.1 Theoretical understanding

Initially, it is feasible to approach the review by synthesizing multiple pieces of evidence extracted from the individual studies included, in order to construct a model that highlights key concepts or issues pertinent to the review question and delineates the relationships among them (POPAY et al., 2006).

The air transport industry, encompassing airlines, airports, and local regulatory agencies, has sought methods to utilize slot allocation coordination as a means to optimize the available airport infrastructure. This approach presents a viable solution to address the rapid increase in air transport demand in the short term. Consequently, two primary questions arise: (i) how can slot distribution be optimized to align with the airport's available infrastructure, and (ii) what are the constituent elements of the airport's operational capacity?

Stakeholders are pursuing a model that facilitates the efficient utilization of airport infrastructure, reduces delays, enhances the probability of airlines retaining allocated slots, and ensures the necessary equity in the distribution of these scarce resources. The long-term impact of such a model includes a more comprehensive understanding of airport limitations, as well as the improvement of the financial performance of all stakeholders and the current slot regulations.

The figure Figure 2.4.provided illustrates the theoretical framework of the problem, progressing from the desired impact and its associated benefits and metrics, to the principal activities required to initiate the model for a more effective resolution of the initial problem (theory of change).

The validity of the initial understanding will be demonstrated within the following sections of the report. The aim is to answer the research questions posed by the systematic literature review

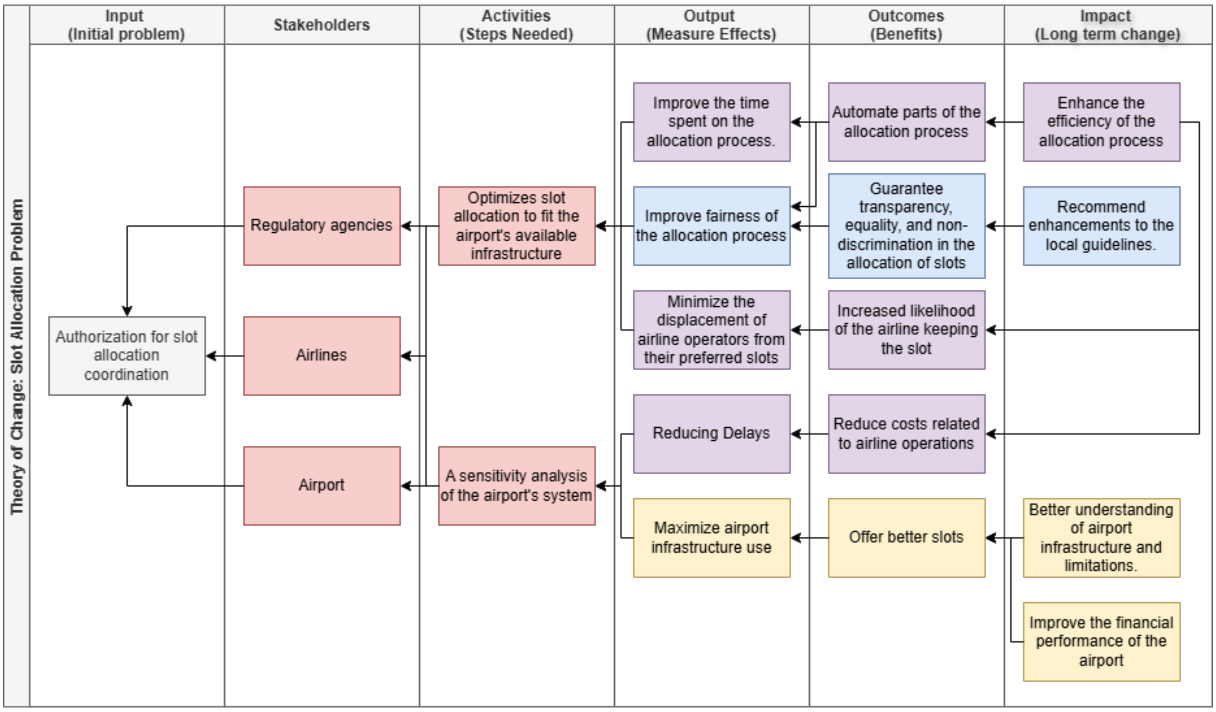


Figure 2.4: Theoretical understanding of the problem: Linking slot optimization and airport sensitivity to enhanced allocation and financial outcomes. Adapted from: (POPAY et al., 2006; EDUSPOTS, 2025).

2.3.2 Preliminary synthesis of findings through tabulation

To synthesize the information derived from the 19 selected studies, the Table 2.3 was constructed. The objective of this table was to systematically present each study, detailing its title, applied methodologies, population under investigation, and principal findings. The numerical designation assigned to each article will serve as a consistent reference throughout this report.

Within the table, in addition to the primary methodologies employed in each study, the applicability of each model was determined based on the coverage classification—namely, whether it pertained to a single airport or an airport network (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017). Furthermore, to facilitate a more comprehensive understanding of the approaches and to underscore the heterogeneity in models, data, and results across the cases, the specific population of each study is also indicated (e.g., the name of the airport or the number of airports comprising the airline network). The main results of each study were explicitly stated in the table, albeit not exhaustively, encompassing both implicit and explicit outcomes and contributions of the respective models.

Table 2.3: Primary studies collection

#	CITATION	TITLE	METHOD	POPULATION	RESULT
1	Zeng et al. (2021) - Research paper	A data-driven flight schedule optimization model considering the uncertainty of operational displacement	Linear programming method (Single airport)	Hangzhou Xiaoshan International Airport (two season)	This research demonstrates a reduction of over 60% in both arrival and departure traffic compared to other optimization model.
2	Ribeiro, Jacquillat & Antunes (2019) - Research paper	A large-scale neighborhood search approach to airport slot allocation	Heuristic solution method and integer programming method (Single airport)	Lisbon Airport (two seasons)	The authors' algorithm produced solutions within 0.1% of the optimum in just a few hours.
3	Zografos, Madas & Androutsopoulos (2017) - Integrated review	Increasing airport capacity utilisation through optimum slot scheduling: review of current developments and identification of future needs	Literature review	N/A	This review encompassed 96 research papers published between 1993 and 2014, developing a classification system for the approaches used in the slot allocation problem.
4	Ribeiro et al. (2018) - Research paper	An optimization approach for airport slot allocation under IATA guidelines	Integer programming method (Single airport)	Madeira and Porto airports (one season)	The model improves slot allocation outcomes by reducing the displacement experienced by airlines by an estimated 4.5% at Madeira and 27% at Porto.
5	Benlic (2018) - Research paper	Heuristic search for allocation of slots at network level	Heuristic solution method (Network of airports)	Set of benchmark (half year)	The article demonstrates that even under strict rules, a network-wide allocation method can achieve a coherent schedule that minimizes delays.
6	Wang et al. (2022) - Research paper	Distribution prediction of strategic flight delays via machine learning methods	Machine learning-based model (Single airport)	Guangzhou Baiyun International Airport (3 years)	Prediction accuracy exceeding 80% was achieved.
7	Jacquillat & Vaze (2018) - Research paper	Interairline equity in airport scheduling interventions	Lexicographic modeling (Single airport)	John F. Kennedy Airport (one calendar day)	The article shows that achieving maximum equity requires no (or minimal) sacrifice in terms of efficiency losses.
8	Zografos & Jiang (2019)	A bi-objective efficiency-fairness model for scheduling slots at congested airports	Bi-objective modelling framework (Single Airport)	Data resembling real world conditions of a Coordinated Airport (one season)	The paper presents a bi-objective model that combines efficiency and fairness. The model uses the ϵ-constraint method, where one objective (fairness) is turned into a constraint and then iteratively tightened to explore trade-offs between fairness and efficiency.
9	Jiang & Zografos (2021)	A decision making framework for incorporating fairness in allocating slots at capacity-constrained airports	Bi-objective modelling framework (Single Airport)	One airport (one season)	The paper presents a framework that combines efficiency and fairness. A key innovation is the incorporation of a voting mechanism. Airlines express their willingness to trade-off efficiency for fairness depending on their individual weight.
10	Jacquillat & Odoni (2018) - Integrated review	A roadmap toward airport demand and capacity management	Literature review	N/A	The review provides a framework that underscores the critical interdependencies between operational/managerial, and economic considerations in airport demand management.
11	Gillen, Jacquillat & Odoni (2016) - Integrated review	Airport demand management: The operations research and economics perspectives and potential synergies]	Literature review	N/A	Introduced an integrated framework and highlighted potential research synergies at the intersection of operations, economics, and management.
12	Dixit et al. (2023) - Research paper	Algorithmic mechanism design for egalitarian and congestion-aware airport slot allocation	A game-theoretic model and a mechanism design solution (Single airport)	Indira Gandhi International Airport and Chennai International Airport (5 calendar days)	The model yielded a 5-20% and 20-30% increase in social utility at DEL and MAA, respectively.
13	Kuran (2020) - Thesis	Heuristic optimization methods for seasonal airport slot allocation	Heuristic solution method (Single airport)	Vienna Airport (5 seasons)	The proposed method demonstrates promising results in minimizing time deviations from the requested slots.
14	Lambelho et al. (2020) - Research paper	Assessing strategic flight schedules at an airport using machine learning-based flight delay and cancellation predictions	Machine learning-based model (Single airport)	London Heathrow Airport (10 seasons)	The proposed prediction models demonstrated high accuracy (79%) across all evaluated approaches. Light Gradient Boosting Machine (LGBM) consistently exhibited superior performance.
15	Keskin & Zografos (2023) - Research paper	Optimal network-wide adjustments of initial airport slot allocations with connectivity and fairness objectives	Two bi-objective models (Network of airports)	16 coordinated and facilitated airports in Brazil (2019 summer season)	The case study results indicate that using connectivity metrics significantly affects the distribution of total displacement among airports.
16	Liu, Zhao & Delahaye (2022) - Research paper	Research on slot allocation for airport network in the presence of uncertainty	Mixed integer programming, (Network of airports)	15 coordinated airports in China (two calendar days)	The model minimizes the sum of displacement costs and delay costs under the worst-case scenario.
17	Wang et al. (2023) - Research paper	Slot allocation for a multiple-airport system considering airspace capacity and flying time uncertainty	Chance constrained programming model (Network of airports)	Five airports within the Guangdong-Hong Kong-Macao Greater Bay Area (one calendar day)	It outperforms those from the certainty model and the original schedule with the cost of a small number of increased slot displacements.
18	Pellegrini et al. (2017) - Research paper	SOSTA: An effective model for the simultaneous optimisation of airport slot allocation	Integer linear programming model (Network of airports)	European Level 2 and Level 3 airports (day with the highest air traffic volume of 2013)	It proved valuable in conducting a comprehensive sensitivity analysis of various model parameters and objective functions.
19	Corolli, Lulli & Ntaimo (2014) - Research paper	The time slot allocation problem under uncertain capacity	Two stochastic programming models (Network of airports)	Multiple European airport network datasets (4 calendar days)	The models can reduce schedule/request discrepancies and operational delays by up to 58%.

2.3.3 Exploring relationships within the data: Concept Mapping

To address the information presented in each of the studies, by exploring their interrelationships through the topics of motivation, applied methods, obtained results, and limitations, thus illustrating the heterogeneity among them, the present systematic literature review developed a conceptual map.

The simplified map, illustrated in the Figure 2.5, presents four subdivisions for enhanced comprehension: (i) Motivation for Improvement, wherein the principal reasons for the development of studies aimed at improving slot allocation at airports will be expounded and interrelated; (ii) Methods, wherein the approaches will be explained, along with their continuous developments and enhancements; (iii) Results, wherein, given the applied methods, the main contributions obtained are presented; and (iv) Limitations, wherein aspects not fully developed within the respective research, those indicated for future work, and gaps in the methodological approaches are elucidated.

The relationships among the studies within each subdivision will be elaborated upon in subsequent subsections. It is pertinent to note that the numerical designation of each study, as presented in the Table 2.3, will be employed to simplify the connections between them.

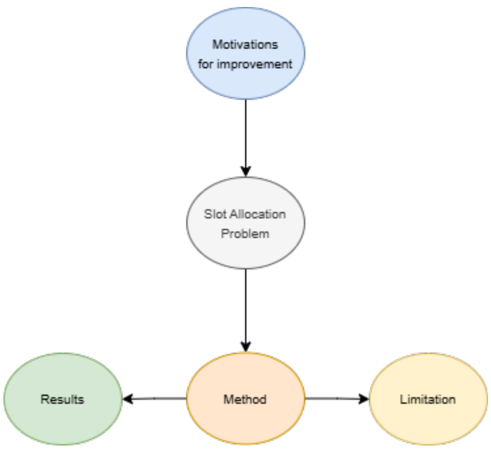


Figure 2.5: Simplified conceptual mapping of the findings from Slot Allocation Problem Systematic Literature Review.

Concept Mapping: Motivations for improvement

Three principal branches originate from the motivations identified among the selected studies: (i) the anticipated increase in air traffic, attributed to the growing number of global passengers; (ii) the balance between efficiency and fairness in slot allocations, representing a pursuit of optimal infrastructure distribution that maximizes its utilization, alongside an approach ensuring that the scarce resource is allocated with transparency, non-discrimination, and equality; and (iii) the inherent complexity of the slot allocation problem, given its combinatorial nature. The Figure 2.6 illustrates the three main branches and their sub-branches, depicting motivations that are a result of the former.

The increasing development of the civil air transport industry is consistently identified across all studies as a primary reason to enhance the efficiency of slot management methods. This factor, combined with the current capacity constraints of airports – whether due to infrastructure limitations or regulations – leads to congestion. This congestion is responsible for widespread delays within airline networks, which can generate costs for all stakeholders in the industry (RIBEIRO et al., 2018; DIXIT et al., 2023).

Prior to the COVID-19 pandemic, global passenger volume was projected to reach 11.4 billion by 2024, representing 124% of the 2019 level. Current projections for 2024 estimate a global passenger volume of 9.5 billion, which is 104% of the 2019 level, indicating a 9% year-over-year growth from the 2023 volume (ACI, 2025). Consequently, despite the impact of the pandemic on the aviation industry, persistent congestion is anticipated at airports worldwide due to this continued growth in demand. This congestion often results in decreased flight punctuality and a diminished passenger experience (LIU; ZHAO; DELAHAYE, 2022).

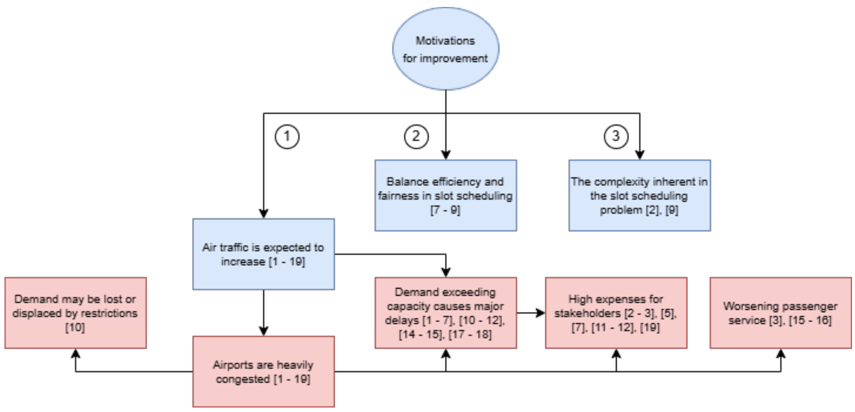


Figure 2.6: Main motivations identified in the studies for addressing the airport slot allocation problem. The articles are referenced by the numbers presented in Table 2.3.

The selected studies highlight the intensifying imbalance between rapidly growing air traffic and limited airspace, leading to reduced flight punctuality and increased delays (COROLLI; LULLI; NTAIMO, 2014; GILLEN; JACQUILLAT; ODONI, 2016; PELLEGRINI et al., 2017; JACQUILLAT; VAZE, 2018; RIBEIRO et al., 2018; JACQUILLAT; ODONI, 2018; ZOGRAFOS; JIANG, 2019; LAMBELHO et al., 2020; ZENG et al., 2021;

KURAN, 2020; KESKIN; ZOGRAFOS, 2023; WANG et al., 2023). This situation underscores the critical need for effective slot allocation strategies to mitigate these adverse effects.

In 2019, the number of flights operated by passenger airlines in mainland China reached 4.611 million, a 6.1% increase compared to the previous year (WANG et al., 2022). These authors, citing CAAC (2019), indicate a trend of increasing air passenger traffic in China. The average delay per passenger flight in 2019 was 14 minutes, representing a slight decrease compared to 2018. Such operational restrictions can lead to lost or displaced demand, negatively impacting the total revenue for stakeholders (JACQUILLAT; ODONI, 2018)

Benlic (2018) shows a projection that indicates that air traffic in Europe will increase to 14.4 million flights by 2035. Consequently, in their most likely scenario, approximately 12% of demand will be unaccommodated, leading to delays.

Corolli, Lulli & Ntaimo (2014) and Zografos, Madas & Androutsopoulos (2017) highlight that the rapid growth of air transport services in Europe, coupled with institutional limitations on the construction of new airports, has resulted in congestion issues manifested by increasing delays. In 2016, the average delay per flight within the European Civil Aviation Conference area was 18 minutes (EUROCONTROL, 2016). Furthermore, the authors note that Air Traffic Management (ATM) inefficiencies in the European Union caused 10.8 million minutes of flight delays in 2012, incurring costs of €4.5 billion for airspace users and €6.7 billion for passengers, and generating 7.8 million tonnes of excess CO2 emissions (IATA, 2014).

Ribeiro et al. (2018) cited data on flight punctuality in the United States and Europe. According to EUROCONTROL (2024), 80.1% of flights arriving in the U.S. were within 15 minutes of their scheduled time, whereas in Europe, this figure was 76.5%.

Ribeiro, Jacquillat & Antunes (2019) assert that the growth in air traffic demand has surpassed the available capacity at numerous airports globally, leading to the frequent occurrence of flight delays and substantial costs for airports, airlines, and passengers. To illustrate this assertion, the authors cite the financial impact of air traffic congestion in the United States, which exceeded $30 billion in 2007 (GILLEN; JACQUILLAT; ODONI, 2016; JACQUILLAT; VAZE, 2018; JACQUILLAT; ODONI, 2018; RIBEIRO; JACQUILLAT; ANTUNES, 2019) and reached $33 billion in 2019 (DIXIT et al., 2023).

Jacquillat & Vaze (2018) and Zografos & Jiang (2019) demonstrated that a primary objective in addressing the slot allocation problem is to balance the preferences and requirements of diverse stakeholders. This often involves navigating trade-offs between efficiency (maximizing the aggregate utility of stakeholders), equity (fairly distributing utilities among stakeholders), and potentially other objectives such as maximizing outcome predictability and ensuring incentive compatibility. This perspective has led authors to propose models incorporating inter-airline fairness measures, which aim to distribute schedule displacement equitably among airlines (FAIRBROTHER; ZOGRAFOS, 2018; JACQUILLAT; ODONI, 2018; ZOGRAFOS; JIANG, 2019; JIANG; ZOGRAFOS, 2021)

These cited imbalances can lead to negative socio-economic and environmental impacts, including increased emissions, and a deterioration of the service quality offered. These consequences affect the travelling public, airlines, airports, and communities situated near airports (KESKIN; ZOGRAFOS, 2023). Furthermore, Zografos, Madas & Androutsopoulos (2017) suggests that objectives could also be established to maximize social benefits.

Another key motivation, as highlighted by Ribeiro, Jacquillat & Antunes (2019), is the limited applicability of exact optimization approaches, which primarily remains confined to small and medium-size airports. This limitation arises from the combinatorial complexity of the problem, rendering it computationally intractable for larger airports. Similarly, Zografos, Madas & Androutsopoulos (2017) supports this view, due to the inherent combinatorial complexity and resource constraints, it is unlikely that a solution algorithm with polynomial time complexity will be found. The complexity of the resulting slot scheduling problem has thus attracted significant interest from the research community, leading to the proposal of several models for optimizing slot allocation decisions (JIANG; ZOGRAFOS, 2021).

Concept Mapping: Methods and Results

To address the research question formulated for this systematic literature review, as detailed in Section 2.1, it is necessary to examine the application of integer programming, queueing theory, and machine learning techniques in addressing the slot allocation problem, consistent with the conceptualization presented in Figure 2.4. The selected studies have identified three primary methodological branches for solving the allocation problem: (i) single airport allocation and (ii) airport network allocation. A further approach involves (iii) reviewing the existing literature and synthesizing the methodologies and results, which constitutes the objective of the current Systematic Literature Review (SLR). Figure 2.7 illustrates the methodologies employed in the reviewed studies alongside their main findings.

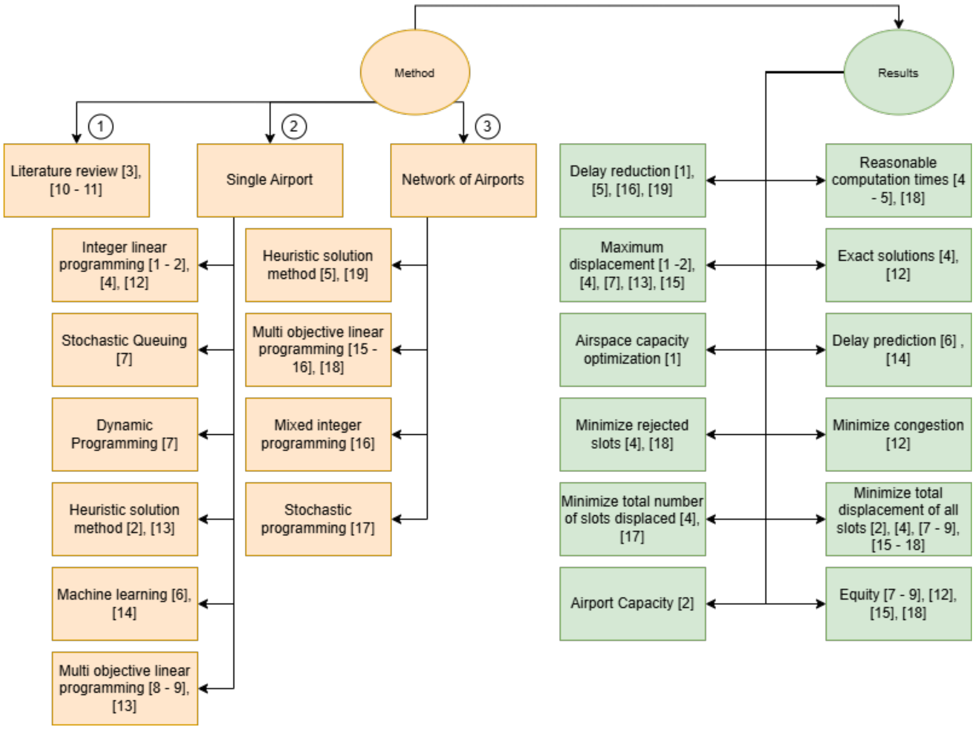


Figure 2.7: Methodological approaches and main findings of reviewed studies on slot allocation. The articles are referenced by the numbers presented in Table 2.3.

Analysis of existing literature review on slot allocation problem

Initially, this report will address the secondary studies that provide a literature review.

Gillen, Jacquillat & Odoni (2016), in their literature review, highlighted the interdependencies among the operating, managerial, and economic dimensions of the slot allocation problem. Their review aimed to identify models that (i) determine appropriate scheduling levels for congestion mitigation and (ii) specify mechanisms for allocating capacity to airlines. The article presented both administrative measures and market-based approaches, explaining and comparing mechanisms such as slot control, congestion pricing, and slot auctions. Gillen, Jacquillat & Odoni (2016) emphasized the inherent uncertainty in addressing delays, meteorological conditions, and the precise timing of movements, which are influenced by various stochastic factors. The authors also explained the complexity of air traffic operations, which can experience multiple bottlenecks across different phases (e.g., departure, en-route, and arrival), each with its own local constraints. Gillen, Jacquillat & Odoni (2016) described models that assess demand performance status but do not focus on optimization. These include deterministic queue models, stochastic queue theory models, and their combinations. To address optimization, the authors cited Integer Programming, which aims to maximize the efficiency of slot allocation. Recognizing the complexity of choosing between price-based and quantity-based instruments, a decision dependent on several market characteristics, the authors established topics to guide the discussion on which method to apply. Finally, Gillen, Jacquillat & Odoni (2016) introduced an integrated framework for airport demand management, outlining the relationships between airport demand, supply, and on-time performance.

Zografos, Madas & Androutsopoulos (2017) aimed to achieve three objectives: (i) a critical review of capacity declaration modeling and optimization strategies for slot allocation within available infrastructure; (ii) an identification of the challenges in existing research and its primary analytical gaps; and (iii) a proposal for future research directions to address the problem. The authors conducted a review of 96 studies, with around two-thirds cited in the article, primarily focusing on slot management approaches rather than market-based methods. The article highlights the necessity of approaches that lead to better quantity (e.g., more slots being accepted than through strict application of IATA guidelines) and better quality (e.g., allocation closer to the initial submission request). It is demonstrated that a significant relationship exists between slot allocation, runway demand, aircraft fleet assignment, airport gate assignment, landing and takeoff sequencing, crew scheduling, as well as ATM ground control or delay programming. Zografos, Madas & Androutsopoulos (2017) also addresses the modeling of airport infrastructure capacity declaration, showcasing various assumed attributes (e.g., level of aggregation, types of movements, time intervals for analysis, rolling constraints, capacity influenced by diverse internal and external factors, weather dependency, etc.). The authors not only explain the different types of approaches in terms of infrastructure (e.g., for a single airport, airport network) but also distribution methods (e.g., deterministic or heuristic). To this end, the formulation of an integer linear programming objective function is presented, within a deterministic allocation model, for both infrastructure categories – single and multiple airports. Finally, the article utilizes the unaddressed or explicitly stated needs for future study developments to generate a classification system for the development of new articles. This classification is based on geographical coverage, number of optimization criteria, type of objective function, nature of capacity, types of constraints, precedence relationships, priority classes, and other regulatory properties.

Jacquillat & Odoni (2018) outline three primary objectives in their study: (i) the estimation of airport capacity and its key drivers; (ii) the presentation of an approach for determining punctuality performance as a function of demand and capacity; and (iii) the introduction of models that could potentially enhance airport demand management mechanisms, summarizing insights and providing a roadmap for future research. The authors define maximum throughput capacity as the average number of aircraft movements that can be processed per unit of time under continuous demand, identifying its main determinants as the number of runways, runway layout and configuration, separation requirements between movements, operational and meteorological conditions, arrival and departure mix, and aircraft mix. These factors directly influence an airport's ability to receive and accommodate air traffic demand. The authors highlight two ways to characterize airport capacity: theoretical, which utilizes the aforementioned data with certain simplifications to calculate the acceptable demand; and empirical, which, through historical data, estimates the Operational Throughput Envelope to characterize the maximum movements of each category that can be accommodated. Regarding airport congestion, the authors categorize models into three types: microscopic, which creates an airport layout and analyzes each aircraft individually, typically involving simulations; mesoscopic, which characterizes runway utilization, taxi-out and taxi-in times, along with other statistical data, to estimate demand; and macroscopic, which uses an aggregated representation of airport operations to generate accurate estimates of delays based on allocated flights and airport capacity. Jacquillat & Odoni (2018) offer insights into the non-linearity between slots, airport capacity, and punctuality performance. They indicate that small increases in slot acceptance can lead to significant negative impacts on punctuality, whereas a more even distribution of flights throughout the day results in lower delay occurrences. Similar to Gillen, Jacquillat & Odoni (2016), the authors suggest that models such as queueing theory can be applied to understand these sensitivities and, consequently, be utilized with market mechanisms to alter the cost per extra movement during peak hours. In conclusion, the authors define a roadmap of actions following traffic forecast definition: airport infrastructure planning, ensuring the configuration is advantageous by avoiding both under-design (leading to lost demand) and over-design (leading to lost profitability); optimization of air traffic regulation and procedures, which implies training and technology application to meet demand safely, or even enable it; configuration of punctuality performance objectives, determining acceptable levels of congestion; and establishment of demand management mechanisms, whether based on administrative or economic principles.

Administrative mechanisms for slot allocation in one airport

In addition to the literature review, primary studies can be categorized into approaches for a single airport or an airport network, resulting in 11 main methods: integer linear programming, stochastic queuing, dynamic programming, heuristic solutions, machine learning, multi-objective linear programming, decision framework, game theory model, congestion-aware egalitarian mechanism, mixed-integer programming, and chance-constrained programming.

As demonstrated in the literature review by Zografos, Madas & Androutsopoulos (2017), a central objective of various approaches is to minimize the deviation between the allocated slot and the airlines' stated need. For approaches applied to a single airport, Zeng et al. (2021) developed an objective function aimed at minimizing this difference and, consequently, improving punctuality and reducing flight delays. This function minimizes the discrepancy between the allocated slot and the actual predicted arrival time, given a probability of achieving that prediction. Regarding constraints, in addition to runway capacity, the model incorporates airspace corridor capacity, the feasible interval for slot allocation adjustments, the maximum turnaround time, and airport apron capacity. The authors applied the model using data from Hangzhou Xiaoshan International Airport (HGH) during the summer and winter seasons of 2017 – and historical data from the same seasons of the previous year. HGH is the ninth most congested airport in China by air traffic movements, with two runways and three airspace corridors for arrivals and departures. Compared to the integer programming model by Zografos, Salouras & Madas (2012) for a single airport – which also considers airport capacity and turnaround time according to IATA guidelines – the model by Zeng et al. (2021) achieved a reduction in peak departure hours and the average departure time during the day of 80% and 75%, respectively. Similar reductions were observed for arrivals, with 60% and 65%, respectively. The difference between departure and arrival delays before optimization was 18 minutes; after optimization, it was reduced to 4 minutes. The authors also analyzed the sensitivity of each constraint on the objective function results.

Dixit et al. (2023) developed a mechanism termed Egalitarian and Congestion Aware Truthful Slot Allocation (ECATS). This method aims to balance several key objectives: efficiency, by allocating slots to those who value them most; egalitarianism, by ensuring fair access for flights connecting remote or less-prioritized cities; and congestion awareness, by preventing over-allocation of flights within specific time slots to mitigate delays and overcrowding. The allocation problem is modeled using an integer programming formulation where the objective function is designed to maximize what the authors term "social utility," a weighted sum incorporating factors such as the remote city opportunity factor (RCOF) and congestion penalties. This approach facilitates a balance between fairness (egalitarianism) and efficiency. The authors applied the ECATS mechanism at Indira Gandhi International Airport and Chennai International Airport, two airports operating at a coordinated level, over five days in January 2020. While acknowledging their lack of access to precise airport capacity information, necessitating an approximation within the model, the authors anticipated that this would not significantly compromise the experimental results. A payment function, based on the principle of marginal contributions and akin to Vickrey-Clarke-Groves (VCG) payments, is integrated to ensure that each airline pays an amount proportional to its impact on the overall outcome. This payment rule incentivizes truthful bidding and prevents airlines from benefiting through misrepresentation of their valuations. The model's performance was evaluated by comparing its efficiency component against the model proposed by Ribeiro et al. (2018) and the existing slot distribution within the test sample, demonstrating superior results for Egalitarian and Congestion Aware Truthful Slot Allocation (ECATS).

Ribeiro et al. (2018) presented a multi-objective optimization model, utilizing a linear programming approach, named the Priority-based slot allocation model (PSAM). The objective function comprises four terms for optimization: the total number of rejected slots, the maximum displacement imposed on any slot, the total displacement across all flights throughout the season, and the total number of displaced slots. Each term is associated with a weight parameter that reflects the priority for optimization, the terms listed above are in their standard priority configuration. The authors also introduced novel constraints that reduce the gap between the integer formulation of the model and its linear relaxation, thereby significantly enhancing its computational performance. This removes the feasible region containing fractional solutions without eliminating any feasible integer solutions, thus ensuring that the optimal integer solution remains unchanged. The article apply the model on the slots requests and allocation data for the Summer Season of 2014 at Madeira and Porto airports in Portugal. It demonstrated exact solutions within reasonable computational times for mid-size airports. Comparisons with slot coordinator decisions indicated that the model accurately captures the main decisions and trade-offs observed in practice. Furthermore, the model improved slot allocation outcomes by reducing the displacement experienced by airlines by an estimated 4.5% and 27% at Madeira and Porto airports, respectively. The authors also provided a sensitivity analysis of various constraints imposed by the IATA guidelines.

Ribeiro, Jacquillat & Antunes (2019) developed a model based on Large-scale Neighborhood Search (LNS), a local search heuristic algorithm, to address the airport slot allocation problem. Building upon their previous work (RIBEIRO et al., 2018), the authors formulated the Priority-based slot allocation model incorporating runway, terminal, and apron constraints (PSAM-RTA), increasing the model's complexity to incorporate terminal and apron capacity constraints. Two heuristics were employed: (i) the first generates a good initial feasible solution rapidly by processing slot requests in decreasing order of frequency, and (ii) the second refines the solution through iterative decomposition of the problem into smaller components and subsequent reoptimization. Using data from Lisbon Airport (LIS) in 2015, the authors found that the constructive heuristic provided feasible solutions faster than CPLEX. Furthermore, the outputs of the constructive heuristic, obtained in less than 30 minutes, outperformed the best solutions achieved with

CPLEX after 2 days, and the improvement heuristic reduced the optimality gap

Kuran (2020) also developed a two-stage heuristic algorithm. The first stage generates a feasible solution by ensuring all capacity constraints are met. The second stage applies a constructive procedure, a heuristic improvement method, focused on assigning requests that could not be accommodated initially to different time slots. The authors applied their model to data from Vienna International Airport (VIE) across five seasons, comparing the results with manually created solutions. The model effectively solved the airport slot allocation problem within short running times, yielding solutions competitive with current operational practices, resulting in low time deviation and minimal fragmentation.

Zografos & Jiang (2019) proposed a bi-objective slot scheduling model aiming to minimize both the total displacement of all slots requested by all airlines and to equalize the average displacement experienced by each airline. The paper offers a comprehensive solution framework employing the ϵ-constraint method, where one objective (fairness) is transformed into a constraint and then iteratively tightened to explore the trade-offs between fairness and efficiency. The presented model was applied to data simulating real-world conditions at a Coordinated Airport. The analysis of the trade-off between the objectives revealed that sacrifices in fairness lead to gains in terms of total displacement. It can be inferred that achieving a fairer scheduling is considerably easier among historic slot requests compared to the scheduling of new entrant and other requests. The efficient frontiers generated by the proposed model can facilitate discussions among various stakeholders, such as airlines, slot coordinators, and airports, to reach a mutually agreed-upon solution. Crucially, these efficient frontiers can also be utilized to avoid dominated (inferior) slot scheduling outcomes.

Building upon the work of Zografos & Jiang (2019), Jiang & Zografos (2021) introduced three objectives related to fairness: (i) minimizing the maximum deviation from absolute fairness, (ii) minimizing the maximum deviation from average fairness, and (iii) minimizing the Gini Index (MGI), adapted to reflect inter-airline fairness in the slot allocation context. Within this framework, for the MGI, the population under consideration comprises the airlines operating at a given airport, and the characteristic examined is the average displacement allocated to each airline. Selecting the most preferable solution from the generated efficient solutions necessitates the involvement of relevant decision-makers/stakeholders, namely airlines and slot coordinators. In the proposed framework, airlines express their preferences regarding the choice of the efficient frontier, while the slot coordinator makes the final selection using the efficient frontier proposed by the airlines. The authors' approach employs a utility function to identify airlines favored or disfavored by different slot allocation outcomes and a voting mechanism to aggregate airline preferences concerning the efficient frontier to be used for solution selection. Subsequently, the coordinator selects the implemented solution by minimizing the weighted normalized distance from the chosen efficient frontier. The proposed decision-making framework was applied to a setting resembling real-world airport demand and supply conditions. The results of the sensitivity analysis indicate that, irrespective of the fairness objective considered, the number of acceptable slot allocation solutions increases with the importance assigned to fairness. Furthermore, the choice of the most preferable acceptable solution exhibits limited sensitivity to the importance assigned by the slot coordinator to the fairness objective.

Jacquillat & Vaze (2018) propose and evaluate a quantitative approach to optimize slot distribution, aiming to achieve on-time performance objectives while minimizing interference with airlines' competitive scheduling and, notably, balancing the impact of such interventions equitably among the airlines. This approach builds upon the Integrated Capacity Utilization and Scheduling Model (ICUSM) framework from Jacquillat & Odoni (2015), incorporating Equity Considerations (ICUSM-E). The modeling framework of the ICUSM integrates an Integer Programming model of scheduling interventions, a Stochastic Queueing Model of airport congestion, and a Dynamic Programming model of airport capacity utilization. The authors introduce the concept of Inter-airline Equity, defined as the ability to balance schedule displacement fairly among airlines. Perfect equity is achieved when the weighted sum of displacements borne by each airline is proportional to the number of flights it has scheduled at the airport. The minimization process is performed lexicographically, first minimizing the largest airline disutility, then the second-largest, and so forth. To achieve this, the authors first establish on-time performance targets, maximizing efficiency, and subsequently optimize equity under these performance and efficiency targets. Jacquillat & Vaze (2018) applied their model to data from September 18, 2007, at John F. Kennedy Airport (JFK). Incorporating inter-airline equity objectives in scheduling interventions for flights with similar valuations can yield significant benefits by distributing scheduling adjustments more fairly among airlines without compromising efficiency. Equity can be attained through comparatively small increases in efficiency, even with substantial differences in flight valuations across airlines and within an airline's own flight portfolio.

As demonstrated in the preceding articles, significant effort has been directed towards optimizing slot allocation, with the primary objective of minimizing displacement to airlines' slot requests. However, slot adherence is susceptible to operational day delays arising from uncertainties such as weather conditions, aircraft maintenance, and passenger-related issues. Lambelho et al. (2020) and Wang et al. (2022) developed a machine learning approach to predict delays and cancellations at a strategic slot allocation phase (six months prior to operation). The authors employed three machine learning algorithms: Multilayer Perceptron (MLP), Light Gradient Boosting Machine (LGBM), and Random Forests (RF). Lambelho et al. (2020) utilized ten strategic flight schedules at London Heathrow Airport (LHR) spanning the period from 2013 to 2018. All classifiers achieved an accuracy of 75% or higher for flight delay classification and 98% or higher for cancellation classification. Given the imbalanced nature of the training and test data, which contained a larger number of not-delayed/not-cancelled flights compared to delayed/cancelled flights, LGBM flight classifiers exhibited the best performance in terms of accuracy, precision, and recall (LAMBELHO et al., 2020). The authors also presented Shapley additive explanation (SHAP) values, illustrating the significant positive or negative impact of specific features on delay/cancellation flight classification and the magnitude of this impact. The features with the highest importance for delay classification were arrival delay, hour of the day, airline type, and aircraft seat capacity. For flight cancellation predictions, the most important features were the origin/destination airport and the operating airline. Wang et al. (2022) tested the algorithms using real data from Guangzhou Baiyun International Airport (CAN), with six flight schedules operated from 2017 to 2020. The results indicated that the accuracy of predicting departure delay at a 65% confidence level and arrival delay at a 50% confidence level could exceed 80%. The authors concluded that MLP exhibited the worst performance, while LGBM and RF showed minimal difference in their results.

Administrative mechanisms for slot allocation in network of airports

As highlighted in the preceding subsection, the current approaches to address or mitigate the slot allocation problem primarily focus on single-airport scenarios. However, the central problem can also be examined from a broader network perspective, where slot allocation impacts multiple airports (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017). A primary limitation of single-airport approach lies in its inability to guarantee the usability of an allocated slot for a specific flight. This stems from the fundamental requirement that a flight necessitates a coordinated pair of slots: one at the origin airport and another at the destination airport (WANG et al., 2023). This Systematic Literature Review (SLR) identified three main methodological categories applied in the analyzed studies: (i) heuristic algorithms, (ii) mixed-integer programming, and (iii) multi-objective linear programming.

Corolli, Lulli & Ntaimo (2014) developed two stochastic programming models for the time slot allocation problem, extending the deterministic single-airport model presented in (ZOGRAFOS; SALOURAS; MADAS, 2012) in two key directions. First, their models simultaneously allocate time slots across multiple airports, explicitly considering –

in addition to other operational constraints – the coherence between the departure time slot at the origin airport and the arrival time slot at the destination airport for each flight. Second, their models incorporate stochastic capacity availability, a factor of particular relevance in strategic planning where uncertainty regarding available resources (i.e., capacity) is exceptionally high. Their stochastic optimization approach yields robust solutions by balancing the immediate costs arising from schedule/request discrepancies with the anticipated future costs of delays associated with a proposed flight schedule on operational days. The authors introduced a feature allowing airport declared capacity to be time-dependent, varying period by period on each operational day. This ensures the fullest and most flexible utilization of limited capacity at congested airports. They presented two alternative formulations: the first, termed "simplified recourse," does not account for the temporal downstream effect of delays, providing a lower-bound estimate of future delays. Its advantage lies in its computability by considering each time instant independently. The second formulation, termed "time-linked recourse," considers delay propagation between consecutive time instants. The authors evaluated their model on a network of 21 European airports, utilizing four distinct datasets across four calendar days. In one test instance, the time-linked recourse formulation, their most precise, enabled a reduction in total delay costs (the sum of schedule/request discrepancies and operational delays) by over 58%. Furthermore, the authors noted that their results could also be interpreted as determining the optimal capacity levels at each considered airport that minimize the sum of schedule/request discrepancies and operational delays. The combination of favorable results and computational viability suggests that the proposed approach is highly promising and warrants further investigation, potentially leading to significant monetary benefits for airlines and other stakeholders.

Benlic (2018) proposed a two-stage heuristic approach. The first stage employs a constructive heuristic procedure to obtain a feasible initial solution by eliminating rotations for which a coherent allocation of slots is not possible. The second stage utilizes an iterative heuristic to optimize the initial solution from the first phase in terms of schedule delay. The authors generated a set of benchmark instances varying in the number of airports within the network and the distribution of requests across these airports. The optimization extends the single-airport slot allocation model presented in (ZOGRAFOS; SALOURAS; MADAS, 2012), where the objective is to maximize the number of accommodated requests while minimizing the absolute difference between requested and allocated times. The results indicated that considering the en-route constraint, inherent to the problem, does not introduce a significant degradation in the schedule compared to the existing practice where slots are allocated independently at each airport.

Liu, Zhao & Delahaye (2022) developed a two-step optimization model to investigate the slot allocation problem within an airport network, considering the impact of uncertainty on airport capacity. The first stage of the model operates at an aggregate level, determining flight numbers within a specific timeframe rather than exact slots for each flight, without requiring precise information on airport capacity. Subsequently, at the individual level, each aircraft is assigned to a slot schedule once exact capacity information becomes available. To facilitate more refined management, the authors employed the Benders decomposition algorithm, supported by the Gurobi solver, which enables the specification of multiple scenarios and the construction of a multi-scenario model based on a single base model. The objective is to minimize the total displacement cost and the maximum displacement cost across the defined set of scenarios, without knowing the probability of each scenario. The first stage involves binary integer programming, while the second stage utilizes multi-scenario programming. The Benders decomposition algorithm treats the first-stage integer programming model as the master model, and the second-stage programming is decomposed into sub-models based on the defined scenarios. For their analysis, the authors selected the 15 busiest airports in China, forming an airport network as the analysis system, and used flight requests from March 1 to March 2, 2019. The results indicate that the model's optimization, by considering more information, reduces the number of flight delays and that the conservatism of the two-stage robust optimization is lower than that of single-stage robust optimization.

Keskin & Zografos (2023) proposed a multi-objective linear programming approach for the simultaneous allocation of slots across a network of airports. The model utilizes the slot allocation outcomes at each individual airport as input for the network-level optimization problem, aligning more closely with the process currently employed by IATA. This approach enables the consideration of local contexts, as well as the incorporation of primary and secondary criteria in the initial slot allocation at individual airports. It then optimally adjusts the resulting individual airport schedules to ensure network-wide flight connectivity by accounting for the interdependencies between flights connecting pairs of airports. To achieve this, Keskin & Zografos (2023) developed bi-objective mathematical models incorporating schedule efficiency (Total displacement and Maximum displacement) and fairness objectives, the latter drawing upon the work of Jiang & Zografos (2021). The authors tested their model on data for coordinated and facilitated airports in Brazil, obtained from the slot coordination database of the Brazilian ANAC. This dataset included 56 national airports, 16 of which are either facilitated or coordinated, during the summer scheduling season of 2019. The case study results demonstrated that employing connectivity metrics significantly affects the distribution of the total displacement among the airports, although the network-wide total displacement itself remains unaffected. The distribution of displacements across airports varies considerably depending on the connectivity metric used. Their analysis regarding inter-flight fairness suggests a weak trade-off between total displacement and inter-flight fairness. This implies that a fairer network-wide schedule can be generated without significantly compromising the efficiency of the allocation.

Pellegrini et al. (2017) developed the Simultaneous Optimization of the airport Slot Allocation (SOSTA) model, a decision support tool designed for the optimal coordination of airport capacity management at the European level. SOSTA aims to optimize slot allocation by (i) penalizing the non-allocation of requested slots, (ii) penalizing the displacement of requests not belonging to a pair of coupled requested slots, and (iii) penalizing the displacement of the requested departure slot and the block time difference resulting from the allocation. The model was tested at Level 2 and Level 3 European airports on June 28, 2013, the day with the highest number of movements in that year. The authors highlighted the difficulty in accessing data on requested airport slots and noted that the available real-world data presented only a small percentage of flight records with discrepancies between requested and allocated slots - this was likely attributed to coordinators having already optimized their flights post-allocation. The results indicated that, across all cases, the proposed displacement differed from the actual displacement by a maximum of 10 minutes. A slightly different combination enabled a reduction in the total displacement by 5 minutes. The final slot allocation led to the saturation of some interval capacity constraints at several airports. In summary, the slot allocation generated by SOSTA closely aligns with the outcome of the actual slot allocation process, suggesting that it effectively reproduces IATA slot regulations and best practices. Pellegrini et al. (2017) also conducted a sensitivity analysis of demand and capacity imbalances and introduced a fairness term for the method.

To allocate airport slots within a Multiple-airport systems (MAS), Wang et al. (2023) proposed a novel uncertainty slot allocation model specifically for an MAS. This model considers fixed capacity constraints alongside the uncertainty associated with flying times between airports and fixes. This uncertainty is formulated using chance constraints, where a violation probability is set as an upper bound, limiting the likelihood of constraint violations. To transform these chance constraints, the authors employed stochastic programming to derive a deterministic model transformation. A scenario generation model, as detailed in Liu, Zhao & Delahaye (2022), is developed to determine flying times and the corresponding probability of each potential scenario. The objective function aims to minimize the total displacement of all flights, thereby reducing the cost of slot interventions, and is formulated as a mixed-integer programming model. The authors applied their model to a multi-airport system in the Guangdong-Hong Kong-Macao Greater Bay Area (GBA), recognized as one of the world's busiest MAS. This area studied comprises three

Level 3 airports: Guangzhou Baiyun International Airport (CAN) and Shenzhen Bao'an International Airport (SZX); two Level 2 airports: Zhuhai Jinwan Airport (ZUH) and Macao International Airport (MFM); and one Level 1 airport: Huizhou Pingtan Airport (HUZ). They utilized the actual flight schedule for a single day (December 21, 2019) as input for their model. The results revealed a negative correlation between the changing rate of total displacements and the violation probability. A smaller violation probability implies stricter constraints, leading to significant changes in allocation results even with small variations in violation probability. Conversely, a larger violation probability entails the consideration of more scenarios, making the total displacements more sensitive to changes in this probability. The results also indicated that displacements were not solely concentrated at the airport with the highest traffic volume. These differences can be partly attributed to the fact that displacements are influenced by both demand and capacity. Furthermore, the study noted that displacements were not evenly distributed across arrival and departure flights. In a deterministic model, while schedules are optimized to meet capacity, there remains a high probability of airspace congestion. In contrast, by considering flying time uncertainty, the likelihood of violating capacity constraints is reduced. Thus, the authors concluded that the robustness of the optimized flight schedule can be substantially improved when allocating airport slots by incorporating flying time uncertainty.

Concept Mapping: Limitations

The articles selected for this Systematic Literature Review (SLR) limitations that can be categorized as follows: (i) Research Gaps, these often stem from assumptions or methodological decisions made during the studies, which, while enabling the pursuit of their research questions, introduce certain caveats; and (ii) Future Work, this encompasses opportunities to expand the research topic and its scope, or to explore subjects not fully addressed in the current articles due to divergence from their primary objectives or a potential for improvement if combined with alternative approaches. Figure 2.8 outlines the primary limitations within each category, emphasizing the relationships among the articles pertaining to this topic.

Figure 2.8: Research gaps and future directions: a synthesis of the selected studies limitations. The articles are referenced by the numbers presented in Table 2.3.

Research Gaps

One prominent research gap identified is the necessity of addressing the complexity of real operational costs. Studies frequently simplify the number of factors analyzed, often due to inherent methodological choices, which directly impacts their results (WANG et al., 2022). For instance, Zografos & Jiang (2019) acknowledged their premise that economic costs associated with flight displacement were uniform across all flights of the same airline and did not vary between airlines. This simplification may not hold true given diverse business models. This issue could potentially be resolved by incorporating airline preferences regarding the allocation of their schedule displacement to various flights at different airports (KESKIN; ZOGRAFOS, 2023; PELLEGRINI et al., 2017). Corolli, Lulli & Ntaimo (2014) further emphasized the need to compare costs of schedule definition in strategic and operational phases, as the former incurs additional crew and passenger-related costs, leading to higher overall expenditures than strategic planning.

A recurring concern in the literature is the challenge of accurately representing real-world airport operational capacity. Ribeiro et al. (2018) emphasized the necessity of capturing additional complexities from IATA guidelines, such as terminal, apron, and noise restrictions, which they subsequently addressed in an extended version of their model (RIBEIRO; JACQUILLAT; ANTUNES, 2019). Zografos, Madas & Androutsopoulos (2017), in their integrated review, elucidated the intricacy of the declared capacity construct. This construct encompasses various attributes, including distributions of arrivals and departures, the number of available slots per time interval, the unit of time, rolling constraints, airport infrastructure, and weather dependency. Achieving exact solutions while increasing capacity constraints to mirror real-world problems, by accounting for the aforementioned characteristics, can substantially increase computational processing time. This makes such approach less favorable for practical application (BENLIC, 2018; WANG et al., 2023).

Other identified gaps include the adaptation to missing initial slot request data (PELLEGRINI et al., 2017; KESKIN; ZOGRAFOS, 2023), as well as the computational performance required for models to provide exact solutions (RIBEIRO et al., 2018).

Future work

The studies selected for this systematic literature review (SLR) consistently include suggestions for future work based on their methods and results. This review has identified eight primary recommendations for further research directions. Some of these have already been addressed in subsequent studies, following the guidance of previous research, and were reviewed in the preceding sections.

The most frequently cited area for future work is the development of heuristic solution algorithms. This is particularly relevant for addressing similar problems at even larger schedule-coordinated airports, as such algorithms could significantly reduce computational costs (RIBEIRO et al., 2018; ZOGRAFOS; JIANG, 2019; LIU; ZHAO; DELAHAYE, 2022). Ribeiro, Jacquillat & Antunes (2019) expanded upon this suggestion by creating a method that combines approximations with exact solutions. Keskin & Zografos (2023) also identified heuristics as a viable alternative for approaching the slot allocation problem within airport networks.

As previously noted, the methodologies for slot allocation can be categorized into approaches for single airports or airport networks. Methods developed for the former could potentially be expanded to address the latter. By integrating various elements, including sector capacity, route priority, and delay propagation effects, the concept of a data-driven model for single airport can also be extended to airport networks (COROLLI; LULLI; NTAIMO, 2014; RIBEIRO; JACQUILLAT; ANTUNES, 2019; ZENG et al., 2021). Benlic (2018), whose research addressed the slot allocation problem through a heuristic algorithm in an airport network context, recommend that network-level approaches should ensure robust algorithmic performance in instances more closely resembling real-world cases, and generalize findings regarding the implications of alternative slot-scheduling policies.

Regarding equity issues in slot distribution, the potential benefits of scheduling interventions also motivate future research directions. There is a recognized need to implement methods that effectively balance fairness and efficiency (WANG et al., 2023). A significant opportunity exists in the design and optimization of scheduling intervention mechanisms that allow airlines to provide their preferred flight schedules (JACQUILLAT; VAZE, 2018; JIANG; ZOGRAFOS, 2021; KESKIN; ZOGRAFOS, 2023). The information shared through such mechanisms should be implemented in a manner that prevents "gaming" by airlines misrepresenting their true information (COROLLI; LULLI; NTAIMO, 2014). Zografos & Jiang (2019) emphasized the necessity of comparing alternative fairness measures proposed in the literature and developing a slot allocation mechanism that incorporates both efficiency and fairness criteria. Another opportunity involves investigating the inclusion of fairness metrics that differentiate between peak and off-peak slot requests, or fairness measures that integrate schedule displacement costs (JIANG; ZOGRAFOS, 2021). Pellegrini et al. (2017) noted that the current slot allocation process, established by IATA, faces criticism for its perceived lack of fairness, barriers to entry (attributed to grandfather rights), and inefficient use of airport capacity, and future work should explore regulation alternatives.

For multi-objective functions, a key area for future work involves the appropriate definition and study of weight factors (COROLLI; LULLI; NTAIMO, 2014). In the longer term, models could address strategic questions related to airport declared capacities, the definition and prioritization of slot allocation objectives, and mechanisms for airport capacity allocation. This would involve accounting for their individual or joint effects on airline schedules, airport operations, and passenger demand. Investigating the efficient trade-off frontier between the objectives considered represents an important avenue for future research (RIBEIRO et al., 2018).

Additional limitations noted for further research and exploration include: demand forecasting, the application period for slot distribution in airport networks, the application of machine learning algorithms for data prediction, fragmentation within the slot allocation system, and the trade-offs between safety and efficiency.

A primary motivation for studying the slot allocation problem stems from the increase in air traffic not being matched by corresponding infrastructure availability. A significant challenge in this context is the forecasting of demand over typical time horizons of thirty to forty years. Consequently, it is imperative that such demand forecasts be developed for a variety of scenarios, reflecting the high variability inherent in the regulatory, competitive, and general economic environment of air transport (JACQUILLAT; VAZE, 2018).

Pellegrini et al. (2017) applied a method to an airport network that incurred high computational costs. They noted that considering only a single day for slot allocation is limiting, given that the slot allocation process prioritizes the assignment of slot series. The authors suggested that the slot allocation problem for an entire season in an airport network, such as in Europe, could be solved heuristically, beginning with the busiest days of the season.

For machine learning models, which utilize historical data to categorize future events, it is believed that prediction performance could be enhanced through the construction of more sophisticated models or the development of more precise loss functions tailored to each algorithm used for the slot allocation problem (ZOGRAFOS; JIANG, 2019). Lambelho et al. (2020) suggested extending the feature set for prediction algorithms to improve accuracy. Future work in this area will evaluate the impact of incorporating flight delay and cancellation predictions into flight scheduling optimization models at the strategic phase.

Fragmentation, defined as the distribution of a set of related slot requests (often termed a series) across different time slots rather than their ideal grouping, essentially quantifies how "spread out" assigned times are for requests that ideally should share a common time slot. This concept applies to airport slot allocation, where the goal is to maintain a consistent and uniform schedule for flights during a season. High fragmentation can lead to delays and inefficiencies. Given its inherent presence in airport operations, the development of strategies to mitigate delays due tofragmentation should continue to be explored (KURAN, 2020).

Liu, Zhao & Delahaye (2022) urged the research community to address its increasing focus on efficiency as the sole objective function criterion, often treating safety capacity merely as a constraint. Consequently, they recommended a more detailed investigation into the trade-off between security and efficiency as dual objectives.

2.3.4 Assessing the robustness of the synthesis: Critical Reflection

With this developed narrative synthesis, the final step of the SLR involves analyzing the robustness of the synthesis itself.

To demonstrate the robustness of this synthesis, it's important to first outline its primary contributions:

1. A transparent and replicable review protocol.
2. An initial insight into the problem, translated into a theoretical understanding.
3. A conceptual map of the state-of-the-art, which revises the key motivations, the methods employed to address the slot allocation problem, their results, and current limitations.

These deliverables offer an understanding of the current status of scientific production focused on enhancing airport efficiency to accommodate the growing demand for air passenger transport. Consequently, this literature review successfully answered the research question, demonstrating how integer programming, queueing theory, and machine learning techniques are utilized. These approaches are directly compared with the methodology outlined by IATA guidelines and various optimization strategies, such as minimizing the disparity between airline requests and offered slots, improving slot distribution, ensuring maximum utilization of airport infrastructure, and reducing costs associated with air network delays.

Consistency is observable among the selected studies on slot distribution, ranging from predicting delays and cancellations using historical data through machine learning to categorize future situations (LAMBELHO et al., 2020; WANG et al., 2022), to methodologies that balance efficiency and fairness in slot distributions through multi-objective integer programming (COROLLI; LULLI; NTAIMO, 2014; JACQUILLAT; VAZE, 2018; JIANG; ZOGRAFOS, 2021; KESKIN; ZOGRAFOS, 2023; RIBEIRO et al., 2018; WANG et al., 2023). Most included studies employed validated quantitative methods, contributing to the internal validity of their respective findings

All articles share a common motivation rooted in the growth of the air transport network for passengers, for which administrative slot management represents a short-to-medium-term solution. Mirroring this motivation, the results also exhibit systemic coherence. The aim is consistently to minimize the disparity between what air operators request and what is allocated at airports, whether or not a fair distribution is sought, in order to reduce delays and cancellations that negatively impact the aviation industry and its stakeholders.

While there is a clear potential for these studies to be applied across various scenarios, populations, and contexts, a notable heterogeneity exists. This heterogeneity is primarily driven by the unique operational characteristics and proprietary data access at different airports, making a direct comparative data synthesis challenging. Consequently, the applicability of specific solutions often remains tied to the characteristics of the airport for which they were developed, suggesting a need for more generalizable frameworks or adaptable models. A benchmarking dataset should be provided to test single and network of airport approaches, which would facilitate statistical analysis of each method through a meta-analysis.

The aforementioned aspects highlight the strengths of this synthesis, several reservations also warrant consideration in this SLR. Firstly, this review does not aim for an exhaustive coverage of all methodologies applicable to the central problem, focusing primarily on the aforementioned methods, some articles cited within the selected studies point to sources that were not collected but should have been included in the review. The omission of certain articles, particularly those exploring optimization of administrative methods, e.g., the Zografos, Salouras & Madas (2012), which is cited by Corolli, Lulli & Ntaimo (2014) and Benlic (2018), might slightly bias the perceived prevalence or effectiveness of the methods identified in this review. Given the context of the research protocol, it's believed that future systematic literature reviews could relax search terms during the planning stage to gather such studies and enhance the review's robustness. Secondly, the observation that queueing theory was primarily mentioned in literature reviews as a sensitivity analysis tool rather than directly applied within the primary studies on slot allocation (unlike integer programming and machine learning) indicates a potential gap in the practical integration of this theoretical approach (GILLEN; JACQUILLAT; ODONI, 2016). This suggests that while recognized for its analytical utility in broader airport models, its direct application to the real-time or strategic complexities of slot management may be underdeveloped or less reported. This gap might represent a frontier for future research, potentially offering new insights into managing real-time slot conflicts and operational delays.

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